

Learning Optimal Liquidation with Closing Auctions

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Abstract

We study an end-of-day liquidation problem in which a trader sells inventory through a central limit-order book before participating in a two-sided closing call. The continuous phase is constrained to one-sided liquidation, whereas the closing-call action set permits signed auction schedules and may therefore result in positive or negative terminal inventory. Whereas standard optimal liquidation models typically rely on terminal inventory penalties to manage end-of-day risk, ignoring the significant liquidity events available in closing auctions, we propose a deep reinforcement learning framework, consisting of a Deep Q-Network and its projected continuous-control actor-critic baselines (DDPG, TD3 and SAC), that explicitly incorporates this mechanism. We introduce an optimal liquidation framework designed to explicitly anticipate the closing auction, continuously refining the projected clearing price as the trading session evolves. We develop a generative stochastic market model to simulate the trading session and to emulate the market. We study the model in a synthetic rough-Heston setting and in a simulation driven by historical S&P 500 mid-price paths, while order flow, auction updates and clearing remain generated by the simulator.

Key words: optimal liquidation, auction trading, reinforcement learning, Deep Q-Learning

1 Introduction

1.1 Reinforcement Learning for Electronic Trading and Liquidation

Market making is a cornerstone of modern electronic financial markets, providing liquidity by continuously posting buy and sell quotes while managing inventory and adverse selection risk with different participants having diverse objectives. Since the seminal work of Avellaneda and Stoikov [4] and its extension to explicit solutions in [31], optimal market making has been studied extensively through stochastic control frameworks, leading to tractable strategies that balance expected profit

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against inventory risk under stylized assumptions on order flow dynamics and price evolution. We refer to [16] for a review of the literature on market making and high-frequency trading and to [5, 6] for recent advances in the topic. These models, however, typically rely on parametric assumptions that are difficult to validate empirically and may fail to adapt to the non-stationarity and strategic complexity of real-world markets.

The rapid growth of electronic trading and data availability combined with AI raising influence in financial industry has motivated the use of reinforcement learning as a flexible, data-driven alternative to classical control methods. Reinforcement learning allows a market maker to learn optimal quoting policies directly from interaction with the market, without requiring full knowledge of the underlying dynamics. Recent studies have demonstrated the promise of RL in market making and related problems, showing improved adaptability to complex market conditions, latent regimes, and evolving order flow patterns. The seminal article [54] introduces a Q-learning algorithm for market making problem on limit order book with many state variables. It has been later extended in for example [7, 55, 59, 27, 30, 36, 41] and adding regret analysis [17, 13]. We also refer to [33, 18, 14] for comprehensive summaries of recent advances in RL techniques in finance.

While the notion of market making is an active topic of research including new developments with reinforcement learning techniques, one aspect remains insufficiently explored in the RL-based market making literature. At the end of the day, most exchanges are closing the market by triggering a closing auction, which plays a fundamental role in price discovery and liquidity provision in modern equity markets. Closing auctions concentrate a significant fraction of daily trading volume and often exhibit dynamics that differ markedly from those observed during continuous trading. Inventory held into the closing auction can be liquidated at a single clearing price, but doing so exposes the market maker to auction-specific price impact, imbalance risk, and strategic interactions. Traditional market making models typically ignore the auction or treat the end-of-day liquidation in an ad-hoc manner, while most reinforcement learning approaches focus exclusively on continuous limit order book trading.

We consider a trader who begins with a positive inventory and submits sell limit orders during a continuous CLOB phase. At the auction opening, the control set is enlarged to signed auction schedules. The agent learns a feature-based policy without knowing the transition law of the exogenous order flow, while the exchange clearing mechanism itself is specified by the model.

1.2 On the Importance of (Closing) Auctions

Early works in financial auctions has been developed in [40]. In this article, Kyle provides the first tractable equilibrium model of a continuous auction with asymmetric information. It explains how prices aggregate private information through order flow, introducing measure of market depth and price impact. This paper became the benchmark for analyzing liquidity, price discovery, and strategic trading in modern auction-based financial markets. While auction markets have been well-investigated in the economical community for discrete-time model, see for example [47, 62, 50]

and have known a growing interest, especially since the work of the Nobel laureate Paul Milgrom, it stays challenging at the high frequency-level and for continuous time framework and has been pointed as one of the most challenging question in financial engineering in [15, Section 5.1].

The big picture of the trading session considered in this paper is the following. During continuous trading, the agent solves a one-sided end-of-day liquidation problem: she can only sell and her inventory remains nonnegative. During the closing call, the admissible action set is enlarged to permit two-sided auction schedules, so that terminal inventory may lie on either side of zero. The auction phase is therefore not constrained to monotone liquidation. Deviations from zero terminal inventory are discouraged through the terminal inventory penalty rather than prohibited by a hard position constraint. At the end of the auction phase, known as the clearing time, a clearing price is set by the exchange to ensure as much transaction as possible to clear the market and trade the asset. This closing auction plays a fundamental role in market liquidation and empowers efficiency of price discovery as explained in [38, 56]. We also refer to [48] for an overview of auction mechanism.

On the one hand, auctions successfully fix mechanical flaws of limit order book like correlation breakdown as explained in [11]. On the other hand, unlike a LOB trading, the key challenge of the auction phase is to set an efficient (clearing) price to trade the asset, given the different operations of market participants, known as the price discovery mechanism [44, 8, 9]. The benefit of auctions for market quality, as reducing the spread between the clearing price and the efficient price of a risk asset has been investigated in the recent literature, for example [21, 37, 19, 58, 34] by proposing incentives and optimal fees scheme to mitigate auctions’ flaws [45, 46].

Finally, we see in many markets nowadays that volumes at the closing auction have increased significantly in recent years. Ignoring this stage is therefore problematic when modeling market making. Furthermore, high-frequency market makers aim at ending each trading day with a flat position: it is likely they will use the auction to close their position if they need to. This underlines the importance of considering closing auctions in an end-of-day market making or liquidation framework.

1.3 Methodology, Contributions and Financial Insights

This work proposes a new end-of-day liquidation and closing-call control model based on a reinforcement learning approach, designed to operate over a typical trading session while explicitly anticipating the closing auction at the end of the session. The goal of this exercise is to study whether different off-policy deep reinforcement learning methods are able to learn stable and effective end-of-day liquidation policies that anticipate a closing auction. We formulate a single-agent reinforcement-learning problem for one-sided CLOB liquidation with an explicitly modeled projected closing-call clearing signal. The proposed framework builds on deep reinforcement learning (DRL). We consider Deep Q-Network (DQN) for the discrete action formulation [64, 51, 23] as the baseline, together with continuous-action actor-critic relaxations (DDPG, TD3, SAC) that lift the discrete quoting grid to a continuous one. These approaches have been successfully applied to a wide range of financial problems, including optimal asset allocation, optimal execution in dark

pools, and market making; see, for instance, [53, 26, 39, 55, 5].

More specifically, we compare a baseline DQN approach for liquidation with auction participation to a continuous-control extension. The goal of Q-learning is to find the optimal action-value function Q which yields the approximately optimized feature-based policy. DQN consists in parameterizing the action-value function with a neural network Q_θ such that optimization over Q becomes optimization over the weights $\theta \in \mathbb{R}^q$, for some $q \in \mathbb{N}^*$. Because the trading session splits into a continuous (CLOB) phase and a closing-auction phase with structurally different state and action spaces, we train two phase-specific networks Q_ϕ and Q_ψ , coupled at the auction opening. The action-value function Q_θ is then trained by minimizing a temporal-difference error on minibatches drawn from an experience-replay memory $\mathcal{D}$ of fixed capacity M , which decorrelates consecutive transitions [51]; training is further stabilized by a target network held fixed and refreshed periodically [60, 52] and by the double- Q correction [61]. To accommodate continuous controls and have more relevant means of comparisons to the DQN method, we additionally consider continuous-action actor-critic methods as a relaxation of the discrete formulation: DDPG [42], TD3 [24] and SAC [32].

For comparison, we use a stylized reference benchmark adapted from the optimal market making model introduced by [4] and derived in [31], which allows us to quantitatively assess the efficiency and performance of the Q-learning-based methods considered in this study. We furthermore compare the performance to the time-weighted average price strategy.

The structure of this work is the following. Section 2 describes the general trading session structure investigated. We first introduce the mathematical framework in Section 2.1, providing rigorous definitions of all stochastic processes, agents, and market participants involved in the model. The continuous trading phase and the auction phase are respectively described in Sections 2.1.1 and 2.1.2. We then introduce the auction clearing mechanism in Section 2.1.3, where we also state a well-posedness result for the stylized clearing equation: the existence of a clearing price under very general supply-demand functions given by Theorem 2.1. Section 2.1.4 proposes an algorithm to predict a hypothetical clearing price during the limit order book trading session and to anticipate the conversion of unmatched limit orders into trade blocks for the future closing auction. At the phase boundary, a calibrated linear representation of the residual exogenous CLOB book is used to initialize the auction book. This algorithm lies at the core of our contribution, as it links the observed residual exogenous CLOB liquidity to the initial auction book. In our algorithm, the agent’s CLOB order is good for one interval and does not carry into the auction. Section 3 defines the controlled simulator configuration, the reduced feature map, admissible controls, the shaped training objective and the economic evaluation criteria. Section 4 introduces both the DQN algorithm and the continuous-control RL methods employed in our framework. Section 5 recalls the main results of [4, 31], which we use as a stylized reference benchmark to compare the profit-and-loss (PnL for short) performance of a market maker who anticipates the closing auction with one who does not. Section 6 explains the numerical methods we considered in this work, that is a generative stochastic model of the market in Section 6.1, and a stylized Heston model in Section 6.3 used to

generate synthetic data for the asset price, combined with limit order book parameters calibrated to reflect the projected future closing auction. Finally, Section 6.4 presents the numerical results obtained when using historical data of S&P 500 assets for the mid price process.

These numerical results highlight the benefits of anticipating the closing auction, as well as the effectiveness of the proposed deep reinforcement learning approach in maximizing the market maker's PnL over a trading session for both stochastic rough models and historical data from the S&P 500.

2 Market Model

2.1 Mathematical Framework and Trading Phases

Along this work, we fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, named the market, where Ω represents all the possible market configurations, $\mathcal{F}$ is a σ -algebra denoting the available information and $\mathbb{P}$ is the market probability. We consider a financial asset traded on the financial market along a trading session with price evolving randomly. We divide the trading period into two phases: a continuous phase on a limit order book with market makers and takers and an auction phase, seen as a usual closing auction. We fix two deterministic times $0 < \tau^{\text{op}} < \tau^{\text{cl}}$ representing respectively the opening and the closing of the auction phase and denote by $h = \tau^{\text{cl}} - \tau^{\text{op}}$ the auction duration. We assume that both $(\tau^{\text{op}}, \tau^{\text{cl}}) \in \mathbb{N}^2$ and h is a positive integer. Therefore, the trading horizon is divided into a continuous phase $[0, \tau^{\text{op}})$, in which the trader interacts with the central limit-order book (CLOB) and a fixed length h closing auction $[\tau^{\text{op}}, \tau^{\text{cl}}]$. The agent acts on a realized decision grid $0 = t_0 < \dots < t_n < \tau^{\text{op}} = t_{n+1} < \dots < t_m < \tau^{\text{cl}} = t_{m+1}$. During the continuous phase, the decision times $t_0, \dots, t_n$ are random times generated by the exogenous market-order processes as in Definition 3.1. Consequently, n is random. The decision clock is nevertheless control-independent: conditional on the same realization of the exogenous market processes, the sequence $(t_i)_{0 \leq i \leq n}$ is the same for every admissible policy. During the auction phase, decisions are made at deterministic one-unit intervals such that $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$, where $t_{n+1+h} = \tau^{\text{cl}}$ is terminal. In particular, $m = n + h$. The initial time t_0 is the start of the considered trading session, but need not be the actual opening time of the market.

We denote by I_t the trader's inventory at time t . This inventory is positive (resp. negative) for long position (resp. short positions) with respect to the traded risky asset.

We denote by $\alpha > 0$ the tick size of the asset, fixed by the exchange. We will consider three types of market participants in this study:

- a strategic trader, called the agent, who liquidates inventory through sell limit orders during continuous trading and submits signed schedules during the closing call,
- exogenous market makers, fixing limit orders and providing liquidity during the continuous trading session on both sides of the LOB and proposing limit prices during the auction session,

- exogenous market takers. These participants submit aggressive market orders during both the CLOB and the auction phases to buy or sell the asset.

While we will focus on the optimization of the agent along the day, we use the term exogenous to emphasize the fact that other market makers and takers' optimizations are not considered here. In fact, we consider a single-agent decision problem against fixed background policies. All “exogenous” market participants are represented by a fixed stochastic data-generating mechanism which will be introduced later. This does not mean these participants have no strategic motives in an actual market, but that their individual optimization problems and endogenous responses to the agent are outside the model. We are therefore solving a best-response problem against the specified background policy, not a market-equilibrium problem. The two trading phases are non-overlapping by construction. We view this as a stylized sequential, Xetra-like chronology rather than as a complete model of a US closing process in which auction order collection may overlap with continuous trading. In the historical-input experiments, US equity data determine only the mid-price paths; order flow, auction updates and clearing remain synthetic. We now turn to the details of the trading period $[0, \tau^{\text{op}}) \cup [\tau^{\text{op}}, \tau^{\text{cl}}]$ composed by continuous trading activities on the CLOB followed by a closing auction.

2.1.1 Trading During the Continuous Phase $[0, \tau^{\text{op}})$

During the continuous phase on the LOB, we assume that each market participant observes the mid price $S_t^{\text{mid}} = \alpha k_t^{\text{mid}}$ at any time $t < \tau^{\text{op}}$, where $k_t^{\text{mid}} \in \mathbb{N}$ represents the number of tick at which the mid price is priced. Exogenous market takers submit market orders on both side of the market and consume liquidity. The number of market takers arriving on the side $\zeta \in \{+, -\}$ follows a counting process denoted by N_t^ζ , where $\zeta = +$ denotes the ask side, and $\zeta = -$ the bid side. In other words, N_t^ζ market takers have arrived on the side ζ (e.g., if $\zeta = +$, then N^+ corresponds to buying market takers, consuming liquidity on the sell side). We assume that each market taker $i \leq N_t^\zeta$ submits a market order with volume $\nu_t^{\zeta, i}$ on the ζ side of the CLOB at time t .

During the continuous phase, the agent acts as a one-sided liquidator and submits only sell limit orders. She submits limit orders characterized by a limit price denoted by $S_t^\bullet = \alpha k_t$ at time t where k_t denoted the number of tick chosen at time t to price the asset, and a proposed volume v_t . The agent therefore submits an order characterized by the pair (k_t, v_t) at time t on the limit order book. We assume that the agent is selling her inventory I on the LOB during the continuous phase. This order is thus a limit order on $v_t \leq I_t$ shares at price $S_t^\bullet = \alpha k_t$.

Remark 2.1. Note that $\delta_t := k_t - k_t^{\text{mid}}$ represents the number of tick between the sell limit order proposed by the agent and the mid price, seen as the ask-spread of the agent.

The liquidity provided by exogenous market makers at price level $S_t^{\zeta, j} = \alpha k_t^{\zeta, j}$, where $k_t^{\zeta, j} = k_t^{\text{mid}} + \zeta j$ for $j \in \mathbb{N}^*$ and $\zeta \in \{+, -\}$ is given by the volume $V_t^{\zeta, j}$ at any time t . The depth of the order book on side ζ is given by

$$L_t^\zeta = \inf\{j \geq 1 : V_t^{\zeta, j} = 0\}.$$

We assume that there exists $L_{\max} > 0$ such that $V_t^{\zeta,j} = 0$ for $j > L_{\max}$.

Assumption 1. *Conditional on the price level selected by the agent, her order has priority over exogenous orders resting at the same price level.*

To abstract from queue-position dynamics and latency competition, we adopted the favorable execution convention that the agent has execution priority with respect to other participants.

Remark 2.2. *Based on the random arrivals of market takers, there is (at least partial) execution on the order of the agent if given that a market taker arrives at time t the following condition is satisfied*

$$\sum_{i=N_t^++1}^{N_t^+} \nu_t^{+,i} > \sum_{j=1}^{\delta_t-1} V_t^{+,j}.$$

Note that as soon as the buying volume of market takers reaches selling index k_t of the agent, her order gets (at least partially) executed.

For $i \in \llbracket 0, n \rrbracket$, let E_{t_i} denote the total number of shares executed over $(t_i, t_{i+1}]$ from the order submitted by the agent at t_i . This quantity is observed at t_{i+1} and is given by

$$\forall i \in \llbracket 0, n \rrbracket, \quad E_{t_i} = \max \left(0, \min \left(v_{t_i}, \sum_{i=N_{t_i}^++1}^{N_{t_{i+1}}^+} \nu_{t_{i+1}}^{+,i} - \sum_{j=1}^{\delta_{t_i}-1} V_{t_i}^{+,j} \right) \right).$$

We recall that E_{t_i} is a random variable since v_{t_i} and $V_{t_i}^{+,j}$ are random. The inventory of the agent between t_i and t_{i+1} is $I_{t_{i+1}} = I_{t_i} - E_{t_i}$.

Motivated by the reinforcement learning approach, we assume that the agent will trade during this session until the deterministic physical time $t_n = \tau^{\text{op}} - 1$ before the market switch to the closing auction phase, where n is the final CLOB decision index; hence $t_0, \dots, t_n$ comprise $n + 1$ CLOB decisions.

Every order submitted by the agent at a CLOB decision time t_i is good for the single decision interval $(t_i, t_{i+1}]$. Any unexecuted remainder expires without a cancellation fee at t_{i+1} . In particular, the order submitted at t_n is active on $(t_n, \tau^{\text{op}}]$ and any remaining quantity expires at τ^{op} . It is not transferred to the auction.

2.1.2 Trading During the Auction Phase $[\tau^{\text{op}}, \tau^{\text{cl}})$

At time τ^{op} , the system transitions to an auction, opened by the exchange. Similarly to [19, 45] and motivated by the reinforcement learning approach with the Markov Decision Process modeling the agent interacting with the market, we assume that the agent is setting bids at deterministic fixed time along the auction duration.

Assumption 2. *The agent bids along the auction at discrete times $\tau^{\text{op}} = t_{n+1} < \dots < t_m < \tau^{\text{cl}}$.*

The inventory $I_{\tau^{\text{op}}}$ of the agent remaining from the continuous phase, *i.e.* that has not been liquidated, is then traded on this auction. More precisely, for all $t \in \{t_{n+1}, \dots, t_m\}$, the agent observes exogenous market and limit orders arriving on the auction. Market orders are composed by a certain volume to be bought/sold no matter the price is set at the clearing time by the exchange, while limit orders are set along the auction through a supply function (functional volume to sell/buy below/above a certain price). Every market participant can cancel prior orders, unlike in the continuous trading phase. The agent chooses an action, which will be a limit order to submit, and/or the cancellation of a previous order. In this sense, the agent reacts to the environment (since she posts her order after seeing the orders of other market participants). After her final action at the terminal trading time $t_m < \tau^{\text{cl}}$, the system transitions to a final state that will allow to compute the clearing price and the exchanged volume, thus allows to compute the terminal reward of the agent. Exogenous market participants do not modify their offers from t_m to τ^{cl} . Solely the agent can cancel his older orders and submit a final limit order. Similarly to [21, 37, 46] we assume that the agent submits a linear supply curve to the auction stated in the following assumption.

Remark 2.3. *The ordering of moves immediately before clearing is a maintained stylization of the model, as the agent has a final reaction opportunity. Numerical conclusions, particularly those concerning late cancellations, are conditional on this convention.*

Assumption 3. *The agent has a linear supply curve $\Sigma_t : p \in \mathbb{R}_+ \mapsto K_t^a(p - S_t^a)$ for all $t \geq \tau^{\text{op}}$, where $K_t^a \geq 0$ and $S_t^a \in \alpha\mathbb{N}$.*

At each $t_j \in \{t_{n+1}, \dots, t_m\}$ the agent controls $K_{t_j}^a \geq 0$ and $S_{t_j}^a \in \alpha\mathbb{N}$ so that $\Sigma_{t_j}(p)$ represents net supply, that is the number of shares the agent is willing to sell at price p . If $\Sigma_{t_j}(p) \leq 0$ the agent is willing to buy at price p or below and conversely if $\Sigma_{t_j}(p) \geq 0$ the agent is willing to sell at price p or above. Allowing both signs enlarges the auction control set beyond monotone liquidation. The objective nevertheless remains end-of-day inventory reduction, enforced through the terminal inventory penalty.

We allow the agent to cancel her past bids at cost $d_j = (j - n - 1)d$ at time t_j for $j \in \{n+1, \dots, m\}$ and $d > 0$. Let $c_{t_j} \in \{0, 1\}$ where $c_{t_j} = 1$ if and only if, at time t_j , the agent cancels all orders submitted prior to time t_j . In particular, at the opening of the auction, we set $c_{t_{n+1}} = 0$. We track these actions through a variable θ_{t_j} , with $\theta_{t_n} = \mathbf{0}$ and $\theta_{t_{n+1}} = \mathbf{0}$, where $\mathbf{0} \in \mathbb{R}^h$ is the $m - n$ -vector with 0 components. We recursively define $\theta_{t_j} = \theta_{t_{j-1}} \vee c_{t_{j-1}} \sum_{k=1}^{j-n-2} \mathbf{e}_k$ for $j \in \llbracket n+2, m+1 \rrbracket$ where $\mathbf{e}_k = (\delta_{i,k})_{1 \leq i \leq m-n} \in \mathbb{R}^{m-n}$. By definition, $\theta_{t_j}^{(s)}$ indicates whether the order submitted at time t_{n+s} for $s \in \{1, \dots, j - n - 2\}$ has been canceled prior to time t_j .

Remark 2.4. *In this formulation, the cancellation penalty $d_j c_{t_j}$ scales linearly with the time index j . We interpret this cost not as a strictly cumulative fee on historical orders – which formally could be subjected to redundant cancellations – but rather as a structural mechanism by the exchange to penalize “illusory liquidity”: it is an economic modeling device designed to discourage repeated provision and withdrawal of liquidity. The “fictive” auction reward introduced below is instead an*

algorithmic shaping term. It is not an exchange rebate and is not included in the economic PnL criterion.

At each trading time $t \in \{t_{n+1}, \dots, t_m\}$, the agent submits an order (K_t^a, S_t^a, c_t) to the market. Note that with the definition above, c_t has no impact on the order (K_t^a, S_t^a) , only on orders sent at time strictly before t . The inventory of the agent remains frozen during the auction, *i.e.* $I_t = I_{\tau^{\text{op}}}$ for $t < \tau^{\text{cl}}$.

During the auction, we suppose in addition that both exogenous market makers and market takers are present in the auction. At each trading time t , the number of bids sent by exogenous market makers is denoted by M_t . Each bid sent by these actors is a limit order, each with volume $g_{i,t}(p)$, which is the supply schedule of the i -th bid offer present at time t for $i \leq M_t$. Each exogenous $g_{r,t}$ is a signed net-supply schedule: positive values represent supply and negative values represent demand.

Market takers submit market orders in the auction. Let N_t^+ (resp. N_t^-) be the number of buying (resp. selling) market orders arrived up to time $t \in \{t_{n+1}, \dots, t_m\}$. For $\zeta \in \{+, -\}$, market taker $i \leq N_t^\zeta$ submits a volume $\nu_t^{\zeta,i}$. Market takers can cancel their order along the auction, for instance, volume $\nu_{t_{n+s}}^{\zeta,i}$ can be set to zero at a time t_j should the market maker of the i -th bid on side ζ decide to cancel his order from time t_{j-1} to t_j , with $i \leq N_{t_{j-1}}^\zeta$, for all $j \in \{n+1, \dots, m\}$ and $s \in \{1, \dots, j-1-n\}$. If no cancellation occurs, we keep $\nu_{t_{j-1}}^{\zeta,i} = \nu_{t_j}^{\zeta,i}$ for $i \leq N_{t_{j-1}}^\zeta$. New market orders having arrived at time t are thus indexed by $N_{t_{j-1}}^\zeta < i \leq N_{t_j}^\zeta$. Note that with this notation, there are indeed more and more market orders in the market over time, as the auction progresses.

2.1.3 Clearing Price Rule and Estimation Along the Auction

After time t_m , the system moves into a final stage to set the clearing price of the auction. The market makers and takers can first send a last order in the auction, then the agent can still send a final limit order and/or cancel past ones. At time $t = \tau^{\text{cl}}$, the auction matches total demand and supply to maximize the exchanged volume at a uniform clearing price $S_{\tau^{\text{cl}}}^{\text{cl}}$. In our stylized continuous clearing mechanism, the uniform clearing price is defined as the zero of aggregate net supply and solves the equation

$$\sum_{i=1}^{M_{t_m}} g_{i,t_m}(p) + \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a(p - S_{t_s}^a) - \sum_{\zeta \in \{+,-\}} \sum_{i=1}^{N_{t_m}^\zeta} \zeta \nu_{t_m}^{\zeta,i} = 0, \text{ for } p \in \mathbb{R}_+. \quad (1)$$

Along the auction, we assume that the agent computes the projected clearing price by solving the equation

$$\sum_{i=1}^{M_{t_{j-1}}} g_{i,t_{j-1}}(p) + \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a(p - S_{t_s}^a) - \sum_{\zeta \in \{+,-\}} \sum_{i=1}^{N_{t_{j-1}}^\zeta} \zeta \nu_{t_{j-1}}^{\zeta,i} = 0, \text{ for } p \in \mathbb{R}_+ \quad (2)$$

which is the clearing price equation were the auction to close at time $t_j \in \{t_{n+2}, \dots, t_{m+1}\}$. The estimation at time $t_j = t_{m+1} = \tau^{\text{cl}}$ corresponds to solve (1) fitting with the exchange clearing rule.

Remark 2.5. For $j \geq n + 2$, the label $H_{t_j}^{\text{cl}}$ denotes the indicative price available to the agent at decision time t_j obtained by solving (2) at time t_j . It is computed from the exogenous auction book recorded at t_{j-1} and from all still-active agent schedules submitted up to and including t_{j-1} . Consequently, the time index of $H_{t_j}^{\text{cl}}$ records when the estimate becomes available, rather than the date of the underlying book snapshot.

We now provide sufficient conditions on ensuring existence of a solution to (2). First, note that Equation (2) can be written as $\Phi_j(p) = 0$ where

$$\Phi_j(p) = \sum_{i=1}^{M_{t_{j-1}}} g_{i,t_{j-1}}(p) + A_j p - B_j - C_j$$

with

$$A_j = \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a, \quad B_j = \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a S_{t_s}^a, \quad C_j = \sum_{\zeta \in \{+, -\}} \sum_{i=1}^{N_{t_{j-1}}^{\zeta}} \zeta \nu_{t_{j-1}}^{\zeta, i}.$$

Theorem 2.1 (Existence of a unique (estimated) clearing price). *Let $t_j \in \{t_{n+2}, \dots, t_{m+1}\}$. Assume that $\lim_{p \rightarrow +\infty} g_{i,t_{j-1}}(p) = +\infty$ and $p \mapsto g_{i,t_{j-1}}(p)$ is continuous and strictly increasing for any $i \leq M_{t_{j-1}}$. Assume moreover $\Phi_j(0) \leq 0$, and the nondegeneracy condition “ $M_{t_{j-1}} \geq 1$ or $A_j > 0$ ” holds. Then, the estimated clearing price equation (2) admits a unique solution.*

Proof. Clearly, Φ_j is continuous, $\lim_{p \rightarrow +\infty} \Phi_j(p) = +\infty$ and for $q > p$,

$$\Phi_j(q) - \Phi_j(p) = \sum_{i=1}^{M_{t_{j-1}}} [g_{i,t_{j-1}}(q) - g_{i,t_{j-1}}(p)] + A_j(q - p),$$

where $A_j \geq 0$. If $M_{t_{j-1}} \geq 1$, the first term is strictly positive, and if $M_{t_{j-1}} = 0$, then $A_j > 0$ and the second term is strictly positive. Hence Φ_j is strictly increasing. The intermediate value theorem gives existence of a zero, and strict monotonicity gives uniqueness. Since $\Phi_j(0) \leq 0$, the zero is nonnegative. $\square$

Theorem 2.1 provides conditions under which the clearing equation in our model admits a unique solution when the background participants' supply and demand schedules are not restricted to be linear. In practice, market participants submit collections of limit orders which the exchange aggregates to reconstruct the supply/demand functions. We refer to [46] for an example. These supply/demand functions need not be linear, and in fact that is not even standard in general. Finally and as we have mentioned earlier, our clearing rule recovers the one stated in [37] or [46] for linear supply and demand curve as stated in the following proposition.

Proposition 2.1. Fix $j \in \{n+2, \dots, m+1\}$. Assume that $g_{i,t}(p) = K_t^i(p - S_t^i)$ with $K_t^i \geq 0$, $\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j > 0$ and $\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j \geq 0$. Then there exists a unique clearing price solving (2) and given by

$$p = \frac{\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j}{\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j} \geq 0.$$

In what follows, we impose the conditions of Theorem 2.1 at each auction evaluation time, so that the estimated and terminal clearing prices are well defined. Precisely:

$$D_j = \sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j > 0 \quad \text{and} \quad F_j = \sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j \geq 0.$$

Note that since prices lie on a grid with tick α , the continuous theoretical root of Equation (2) is rounded to the closest tick. More precisely, if its solution is p_j^* , the numerical indicative price is $p_\alpha = \alpha \lfloor p_j^* / \alpha + 1/2 \rfloor$. This creates a subsequent absolute aggregate imbalance of at most $D_j \alpha / 2$, which we solve numerically with a stylized pro-rata allocation rule.

The agent's requested signed quantity at the rounded clearing price

$$\tilde{Z}_{\tau^{\text{cl}}} = \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a \left(S_{\tau^{\text{cl}}}^{\text{cl}} - S_{t_s}^a\right).$$

Let

$$\forall p \in \mathbb{R}_+, \quad \Sigma_{\tau^{\text{cl}}}^a(p) = \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a (p - S_{t_s}^a).$$

At the rounded terminal price p^α , total supply and demand are

$$Q^S(p^\alpha) = \sum_{i=1}^{M_{t_m}} g_{i,t_m}(p^\alpha)_+ + \Sigma_{\tau^{\text{cl}}}^a(p^\alpha)_+ + \sum_{i=1}^{N_{t_m}^-} \nu_{t_m}^{-,i},$$

and

$$Q^D(p^\alpha) = \sum_{i=1}^{M_{t_m}} (-g_{i,t_m}(p^\alpha))_+ + (-\Sigma_{\tau^{\text{cl}}}^a(p^\alpha))_+ + \sum_{i=1}^{N_{t_m}^+} \nu_{t_m}^{+,i}.$$

Let

$$\rho_S = \begin{cases} 1 & \text{if } Q^S = 0 \text{ or } Q^S \leq Q^D \\ Q^D / Q^S & \text{if } Q^S > Q^D \end{cases}$$

and

$$\rho_D = \begin{cases} 1, & \text{if } Q^D = 0 \text{ or } Q^D \leq Q^S \\ Q^S / Q^D & \text{if } Q^D > Q^S \end{cases}.$$

The actual executed volume of the agent at the clearing time is then given by

$$Z_{\tau^{\text{cl}}} = \rho_{\tau^{\text{cl}}}^S[\tilde{Z}_{\tau^{\text{cl}}}]_+ - \rho_{\tau^{\text{cl}}}^D[-\tilde{Z}_{\tau^{\text{cl}}}]_+.$$

The agent's active schedules are aggregated into a single net schedule before this allocation, so no self-trade is counted. A negative terminal inventory is an admissible outcome of the enlarged auction action set rather than a violation of an inventory constraint. Deviations from zero are discouraged through the terminal inventory penalty.

2.1.4 Projected Hypothetical Clearing Price During the Continuous Session

During the continuous phase $[0, \tau^{\text{op}})$, we assume that the agent is estimating the clearing price of the auction. For that purpose, the agent observes all outstanding (unexecuted) limit orders and treats them as if they were submitted to a fictitious auction, where they would be jointly matched to infer the implied clearing price. The agent is then creating a projected hypothetical clearing price H_t^{cl} along the duration of the LOB trading before the closing auction starts. The computation is detailed in Algorithm 1, the calibration of the different characteristics of the auction prices and parameters comes from [37].

At t_i , the previous strategic CLOB order has already expired. Therefore, O_i denotes the standing exogenous CLOB multiset observed before the action at t_i .

Algorithm 1 Computation of $H_{t_i}^{\text{cl}}$ for $i \in \llbracket 0, n \rrbracket$

Require: Tick size $\alpha > 0$, smoothing parameter $\eta_H \in (0, 1]$, initial value H_0

```
1: Initialize  $H_0^{\text{cl}} \leftarrow H_0$ 
2: for  $i = 1, \dots, n$  do
3:   Record standing orders before time  $t_i$  by multiset  $O_i \in \mathcal{M}_f(\alpha\mathbb{N} \times \mathbb{R}_+)$   $\{\mathcal{M}_f(\cdot)$  denotes the
   finite multisets. $\}$ 
4:   if  $O_i \neq \emptyset$  then
5:      $\mathcal{K}_i \leftarrow \text{proj}_1(O_i)/\alpha$  {Standing price levels}
6:     Register  $Q_i(k) = \sum_{(p,v) \in O_i} v \delta_{p, \alpha k}$  for  $k \in \mathcal{K}_i$  and  $Q_i(k) = 0$  if  $k \notin \mathcal{K}_i$ 
7:     for  $k \in \mathcal{K}_i$  do
8:        $\hat{e}_i^k \leftarrow \frac{1}{i} \sum_{s=1}^i Q_s(k)$  {Average volume available at level  $k$ , with  $\hat{e}_i^k > 0$  for  $k \in \mathcal{K}_i$ }
9:        $\hat{\zeta}_i^k \leftarrow \frac{1}{i} \sum_{s=1}^i Q_s(k)^2$  {Average squared volume available at level  $k$ }
10:       $\hat{K}_i^k \leftarrow \max(0, (2\hat{e}_i^k - \hat{\zeta}_i^k / \hat{e}_i^k) \alpha^{-1})$  {Calibrated slope at level  $k$ }
11:    end for
12:    if  $\sum_{k \in \mathcal{K}_i} \hat{K}_i^k > 0$  then
13:      Solve  $\sum_{k \in \mathcal{K}_i} \hat{K}_i^k (\alpha k - p) = 0$  for  $p$  and denote the solution  $\tilde{S}_{t_i}$  {Clearing price rule}
14:       $H_{t_i}^{\text{cl}} \leftarrow H_{t_{i-1}}^{\text{cl}} + \eta_H (\tilde{S}_{t_i} - H_{t_{i-1}}^{\text{cl}})$  {Smoothed update rule}
15:    else
16:       $H_{t_i}^{\text{cl}} \leftarrow H_{t_{i-1}}^{\text{cl}}$ 
17:    end if
18:  else
19:     $H_{t_i}^{\text{cl}} \leftarrow H_{t_{i-1}}^{\text{cl}}$ 
20:  end if
21: end for
```

Finally, the agent is penalized for submitting orders below the clearing price estimate $H_{t_i}^{\text{cl}}$. This is because an order with a price below the hypothetical clearing price would tell the agent to rather wait for the auction to liquidate her shares. This penalty will be detailed in the next section as a penalization for the reward function.

Let $O_{\tau^{\text{op}}-}^*$ denote the multiset of exogenous CLOB orders remaining after all executions on $(t_n, \tau^{\text{op}}]$. The unexecuted remainder of the agent's order submitted at t_n expires at τ^{op} and is excluded from this multiset. To map the residual CLOB book into the linear-schedule representation used in the auction, we repeat the final calibration step of Algorithm 1 with the residual multiset $O_{\tau^{\text{op}}-}^*$ in place of the pre-action multiset O_n , while the earlier snapshots $O_1, \dots, O_{n-1}$ are left unchanged. Denote the resulting slopes by $\hat{K}_\star^k$ and let

$$\mathcal{K}_\star = \left\{ k : O_{\tau^{\text{op}}-}^* \text{ contains positive volume at } \alpha k \text{ and } \hat{K}_\star^k > 0 \right\}.$$

At the auction opening, we assume that the initial exogenous auction book contains the carry-over schedules $g_{k, \tau^{\text{op}}}^\star(p) = \hat{K}_\star^k(p - \alpha k)$ for $k \in \mathcal{K}_\star$. These schedules are included in $M_{\tau^{\text{op}}}$ before any auction-specific arrivals are processed. Let $D^\star = \sum_{k \in \mathcal{K}_\star} \hat{K}_\star^k$ be the aggregate carry-over slope. If $D^\star < D_\mu$, for some numerical lower slope bound $D_\mu > 0$, we add the “persistent” fallback schedule $g^0(p) = (D_\mu - D^\star)(p - S_{\tau^{\text{op}}}^{\text{mid}})$ which cannot be canceled. If $D^\star \geq D_\mu$, one carry-over schedule with

maximal slope is designated as “persistent”.

Remark 2.6. *This construction is a stylized linear representation of the residual exogenous CLOB book. It does not preserve the original side of each individual CLOB order. Instead, each retained price level becomes the zero of a two-sided linear net-supply schedule centered at that level.*

3 Learning Problem for Optimal Liquidation with Closing Auction

We now turn to the discretization of the problem.

Definition 3.1. *Set $t_0 = 0$. For every $i \geq 0$ such that $t_i < \tau^{\text{op}} - 1$ and every $\zeta \in \{+, -\}$, define $\tau_i^\zeta = \inf\{s > t_i : N_s^\zeta > N_{t_i}^\zeta\}$ and $\tau_i := \tau_i^+ \vee \tau_i^-$ with the convention $\inf \emptyset = +\infty$. We set $t_{i+1} = (\tau^{\text{op}} - 1) \wedge [\tau_i \vee (1 + \lfloor t_i \rfloor)]$ until the bound $\tau^{\text{op}} - 1$ is attained. The final CLOB decision index is $n = \inf\{i \geq 0 : t_i = \tau^{\text{op}} - 1\}$. Thus there are $n + 1$ CLOB decisions, $t_0, \dots, t_n$, and we set $t_{n+1} = \tau^{\text{op}}$ and $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$. Finally, $m = n + h$, so that $t_{m+1} = \tau^{\text{cl}}$, and $\mathcal{T} = (t_i)_{0 \leq i \leq m+1}$.*

Remark 3.1. *Except for the final capped interval ending at $\tau^{\text{op}} - 1$, the maximum $\tau_i^+ \vee \tau_i^-$ means that a new decision occurs only after both sides have received at least one additional market order. For simulation, the realized grid is a functional of the exogenous order-flow path. At decision i , however, only $t_0, \dots, t_i$ and information available at t_i are revealed to the agent. The last CLOB decision occurs at the deterministic physical time $t_n = \tau^{\text{op}} - 1$, while the random index n determines how many earlier CLOB decisions occurred.*

In the following, i denotes the decision index whereas t_i denotes the corresponding physical time. We do not identify these two quantities during the continuous phase, since $(t_i)_{0 \leq i \leq n}$ is random. Note however that during the auction phase the two are related explicitly by $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$. For simplicity, albeit an abuse of notation, when there is no ambiguity, we write a generic decision time as t .

The latent state is denoted by X and defined for any time $t \in \mathcal{T}$ as a tuple

$$X_t = (t, X_t^1, X_t^2, X_t^3, X_t^4, X_t^5, X_t^6, X_t^7, X_t^8, X_t^9, X_t^{10}, X_t^{11}, X_t^{12}, X_t^{13}, X_t^{14}, X_t^{15}, X_t^{16}, X_t^{17}, X_t^{18}),$$

where each attribute encodes one of the market characteristics before the choice of an action by the agent as detailed below.

State Variables

- Inventory: $X_t^1 = I_t$ represents the inventory of the market maker at time t : per construction, we restrict $I_t \geq 0$ for $t < \tau^{\text{op}}$ and I_t frozen for $\tau^{\text{op}} \leq t < \tau^{\text{cl}}$ with $I_{\tau^{\text{cl}}} \in \mathbb{R}$;
- Volume executed at the clearing: $X_t^2 = 0$ for $t < \tau^{\text{cl}}$ with $X_{\tau^{\text{cl}}}^2 = Z_{\tau^{\text{cl}}}$ and $H_{\tau^{\text{cl}}}^{\text{cl}} = S_{\tau^{\text{cl}}}^{\text{cl}}$;

- Hypothetical/estimated auction's clearing price: $X_t^3 = H_t^{\text{cl}}$ represents the hypothetical clearing price as defined in Section 2.1.4 during the continuous trading phase for $t < \tau^{\text{op}}$ or the estimated clearing price as defined in Section 2.1.3 as the solution to Equation (2) during the auction trading phase for $\tau^{\text{op}} < t < \tau^{\text{cl}}$ and $H_{\tau^{\text{op}}}^{\text{cl}} = H_{t_n}^{\text{cl}}$;
- Limit order book depth: $X_t^4 = L_t^+ \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$ and $X_t^5 = L_t^- \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$ represent respectively the depth in the limit order book on the ask (resp. bid) side;
- Number of limit order in the auction: $X_t^6 = M_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$;
- Number of investors in the auction: $X_t^7 = N_t^+ \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$ and $X_t^8 = N_t^- \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$ represents respectively the number of aggressive order sent in the auction to buy (resp. sell) the asset;
- Cancellation history: $X_t^9 = \theta_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$ represent the vector of canceled orders in the auction up to time t ;
- Mid price: $X_t^{10} = S_t^{\text{mid}}$
- Volume sent by investors in the auction: $X_t^{11} = (\nu_t^{+,i} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}})_{1 \leq i \leq N_t^+}$ and $X_t^{12} = (\nu_t^{-,i} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}})_{1 \leq i \leq N_t^-}$ represent the number of aggressive orders sent in the auction to buy (resp. sell) the asset
- Volume in the limit order book (ask/bid side): $X_t^{13} = (V_t^{+,j} \mathbf{1}_{\{t \leq \tau^{\text{op}} - 1\}})_{1 \leq j \leq L_t^+}$ and $X_t^{14} = (V_t^{-,j} \mathbf{1}_{\{t \leq \tau^{\text{op}} - 1\}})_{1 \leq j \leq L_t^-}$ are the volume existing in the limit order book on the ask and bid side at any depth
- Limit order in the auction: $X_t^{15} = (g_{i,t})_{1 \leq i \leq M_t}$, with $X_t^{15} = \emptyset$ before the auction;
- Price history in the auction: $X_t^{16} = S^a(t)$ where $S^a(t) := (S_{t_{n+1}}^a, \dots, S_{t-1}^a, 0, \dots, 0)$ represents the vector of limit order prices submitted in the auction up to time t ;
- Supply/Demand slope history: $X_t^{17} = K^a(t)$ where $K^a(t) := (K_{t_{n+1}}^a, \dots, K_{t-1}^a, 0, \dots, 0)$ represents the slope of the limit order submitted up to time t ;
- Decision index: $X_t^{18} = i$ if $t = t_i$ is the time of the i -th decision: let us denote i_t the decision index of time t for a generic time $t \in \mathcal{T}$.

Remark 3.2. In the linear specification of Proposition 2.1 and in the numerical simulator, $g_{i,t}(p) = K_t^i(p - S_t^i)$ so X_t^{15} can equivalently be represented by $((K_t^i, S_t^i))_{1 \leq i \leq M_t}$.

Remark 3.3. Note that X^{16}, X^{17} are vectors of size $h = m - n$ with 0 components after time t .

The state vector X_t is viewed as an element of the state space $\mathcal{X}$ induced by the above variables, that is

$$\mathcal{X} = [0, \tau^{\text{cl}}] \times \mathbb{R}^2 \times \mathbb{R}_+ \times \mathbb{N}^5 \times \{0, 1\}^h \times \alpha\mathbb{N} \times \Psi(\mathbb{R}_+)^4 \times \Psi(\mathcal{G}) \times (\alpha\mathbb{N})^h \times \mathbb{R}_+^h \times \mathbb{N}$$

where

$$\mathcal{G} = \left\{ g \in \mathcal{C}(\mathbb{R}_+) : g \text{ is strictly increasing, } \lim_{p \rightarrow +\infty} g(p) = +\infty \right\}$$

is the space of functions that satisfy the assumptions of Theorem 2.1, and for a set E , the space $\Psi(E)$ is defined as

$$\Psi(E) = \bigsqcup_{k=0}^{+\infty} E^k$$

with $E^0 = \{\emptyset\}$.

Remark 3.4. *While exchanges like the NYSE usually only display statistics on the total imbalance between buy and sell orders combined with an indicative price, it does not display all the orders that have been submitted at each price.*

Theorem 2.1 permits general exogenous schedules. From this point onward, the learning problem and the numerical simulator use the linear specialization of Proposition 2.1 and Remark 3.2. Accordingly, the aggregate slope and intercept features used below are well defined. We assume that the simulator records the complete latent configuration X_t , while the learning algorithms receive only the reduced feature vector $\varphi(X_t)$. We assume that these aggregate features are available to the agent. This reduced state is $\varphi(X_t)$ with

$$\begin{aligned} \varphi(X_t) = & (t, I_t, H_t^{\text{cl}}, S_t^{\text{mid}}, i_t, L_t^+ \mathbf{1}_{\{t < \tau^{\text{op}}\}}, L_t^- \mathbf{1}_{\{t < \tau^{\text{op}}\}}, V_t^{+,1} \mathbf{1}_{\{t < \tau^{\text{op}}\}}, V_t^{-,1} \mathbf{1}_{\{t < \tau^{\text{op}}\}}, \\ & M_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, N_t^+ \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, N_t^- \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, C(X_t) \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, A_{i_t} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, B_{i_t} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, \\ & G_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, \mathcal{I}_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, J_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}) \end{aligned}$$

with A_{i_t} and B_{i_t} given by the expressions of A_j and B_j in Section 2.1.3, $C(x) = \max_{1 \leq i \leq m-n} \{(1 - x_i^9) \mathbf{1}_{\{x_i^{17} > 0\}}\}$ for a state $x \in \mathcal{X}$, and

$$G_t = \sum_{i=1}^{M_t} K_t^i, \quad J_t = \sum_{i=1}^{M_t} K_t^i S_t^i, \quad \mathcal{I}_t = \sum_{i=1}^{N_t^+} \nu_t^{+,i} - \sum_{i=1}^{N_t^-} \nu_t^{-,i}.$$

Let $\mathcal{Y} = \varphi(\mathcal{X})$.

Action Variables We now turn to the actions of the strategic market maker. Given a state vector $X_t \in \mathcal{X}$, we define the action vector A_t as $A_t = (A_t^1, A_t^2, A_t^3, A_t^4, A_t^5)$ where each component represents a particular action.

- Volume set in the limit order: $A_t^1 = v_t \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$;
- Depth offset in the limit order book: $A_t^2 = \delta_t \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$;
- Supply/demand slope and reference price: $A_t^3 = K_t^a \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$; and $A_t^4 = b_t^a \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$ such that $S_t^a = S_{\tau^{\text{op}}}^{\text{mid}} + \alpha b_t^a$;

- Order cancellation in the auction: $A_t^5 = c_t \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$.

We view such actions as part of the ambient action set

$$\mathcal{A} = \llbracket 0, V_{\max} \rrbracket \times \llbracket 0, L_{\max} \rrbracket \times \beta \llbracket 0, K_{\max} \rrbracket \times \llbracket -B_{\max}, B_{\max} \rrbracket \times \{0, 1\}$$

where $V_{\max}, L_{\max}, B_{\max}, K_{\max} \in \mathbb{N}^*$ are numerical bounds defining the finite action grid. In particular, $B_{\max}$ is a numerical truncation parameter chosen sufficiently large for the simulated price range.

Recalling that on the limit order book the market maker is liquidating his inventory, hence submits a volume $A_t^1 \leq I_t = X_t^1$, at a price $S_t^{\text{mid}} + \alpha A_t^2 \geq S_t^{\text{mid}}$. During the auction phase, the market makers can cancel previous orders exactly once, thus $A_t^5 = c_t \leq \max_{1 \leq i \leq t - \tau^{\text{op}}} (1 - \theta_t^{(i)}) =: C_t$ with the convention that $\max_{\emptyset} = 0$ (for $t \leq \tau^{\text{op}}$).

Definition 3.2 (Admissible actions). *Given an observed state $y \in \mathcal{Y}$, the set $\text{Adm}(y)$ of admissible actions is defined as*

$$\begin{aligned} \text{Adm}(y) = \{a \in \mathcal{A} : a_1 \leq y_2, a_5 \leq y_{13}, y_4 + \alpha a_4 \geq 0, a_1 = 0 \implies a_2 = 0, \\ a_3 = 0 \implies a_4 = 0, (a_1, a_2) \mathbf{1}_{\{y_1 \geq \tau^{\text{op}}\}} = 0, (a_3, a_4, a_5) \mathbf{1}_{\{y_1 < \tau^{\text{op}}\}} = 0\} \end{aligned}$$

where the last two components embed the phase compatibility of admissible actions.

Remark 3.5. *Note that S_t^{mid} is constant during the auction. Furthermore, since all X_t^1 , X_t^{10} and $C(X_t)$ are encoded in $\varphi(X_t)$, all admissible actions for X_t are exactly those admissible for $\varphi(X_t)$.*

Definition 3.3 (Admissible policies). *An admissible policy is a measurable map*

$$\pi : y \in \mathcal{Y} \longmapsto \pi(\cdot \mid y) \in \mathcal{P}(\mathcal{A})$$

such that $\pi(\text{Adm}(y) \mid y) = 1$ for all $y \in \mathcal{Y}$. We denote Π the set of these admissible policies.

Remark 3.6. *We use the word configuration for X_t and feature vector for $\varphi(X_t)$. Neither process is claimed to be a finite-dimensional Markov state. The algorithms are feature-based numerical approximations for the specified controlled simulator, and no global optimality claim is made.*

Reward We define the reward on three regions as explained below.

1. During the continuous trading session for $t < \tau^{\text{op}}$. The market maker submit a price $S_t^\bullet = S_t^{\text{mid}} + \alpha A_t^2$. The volume executed is given by E_t . The cash proceeds are thus given by $S_t^\bullet \times E_t$. We moreover assume that the market maker penalizes the execution by comparing the price executed with the hypothetical clearing price $H_t^{\text{cl}} = X_t^3$. If $S_t^\bullet > H_t^{\text{cl}}$, the market maker receives the full profit otherwise if $S_t^\bullet < H_t^{\text{cl}}$ the market maker may regret the execution. We

assume that the difference between X^3 and $S^\bullet$ tolerated is given by $k^\star\alpha$ for some $k^\star$ fixed. It means that as soon as

$$(H_t^{\text{cl}} - S_t^\bullet)_+ \leq k^\star\alpha,$$

the market maker still get a profit from the execution on the limit order. We thus introduce a penalty function $f^c : \mathbb{R} \rightarrow [0, 1]$ continuous and increasing such that f^c is zero on $\mathbb{R}_-$ and $f^c(k^\star\alpha) = 1$, such that the reward of the market maker is given by

$$r_t(X_t, A_t) = S_t^\bullet E_t f^c(k^\star\alpha - (H_t^{\text{cl}} - S_t^\bullet)_+).$$

2. During the auction trading session for $\tau^{\text{op}} \leq t < \tau^{\text{cl}}$. The market maker submits a slope $K_t^a = A_t^3$ and a price $S_t^a = S_{\tau^{\text{op}}}^{\text{mid}} + \alpha A_t^4$. The agent receives a “fictive” reward $K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a)$, where H_t^{cl} is the anticipated clearing price (were the auction to close at time t). The agent is penalized for canceling previous orders at cost d_t at time t , yielding a penalty $-d_t c_t$. Here, d_t is an exchange cancellation fee. Because the auction action set is two-sided, a submitted schedule may imply a projected purchase at the indicative price. This happens when $H_t^{\text{cl}} \leq S_t^a$: the market maker is willing to buy $K_t^a (S_t^a - H_t^{\text{cl}})$ shares at price H_t^{cl} or below. To retain a liquidation preference in the training signal, we optionally attenuate the negative shaping term associated with such projected purchases. We hence introduce a function $f^a : \mathbb{R} \rightarrow \mathbb{R}_+$ convex, continuous and decreasing such that f^a is zero on $\mathbb{R}_+$ and $f^a(x) \leq (-x)_+$. The term $f^a(K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a))$ attenuates or even removes the adverse consequence of buying. This attenuation is neither an admissibility constraint nor an economic cash flow. The reward of the market maker is given by

$$r_t(X_t, A_t) = K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a) + f^a(K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a)) - d_t c_t.$$

The “fictive” reward during the auction is a training-shaping term designed to provide a denser learning signal for actions that improve the agent’s position relative to the current indicative clearing price. We consider here that the fictive reward is not restored in hindsight after cancellation of an order. Moreover, to retain a liquidation preference while permitting two-sided auction actions, we optionally attenuate the shaping reward associated with projected purchases. This term is a modeling choice in the training signal and is neither an admissibility constraint nor an economic cash flow.

3. Final reward at the clearing for $t = \tau^{\text{cl}}$. At the clearing time, the clearing price $S_{\tau^{\text{cl}}}^{\text{cl}}$ is determined and orders are matched. The market maker makes the profit or loss $S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}}$ based on the orders she sent to the market and did not cancel by the clearing time. The agent is furthermore penalized for holding inventory. We introduce $\lambda > 0$ as a penalization parameter. Furthermore, the agent’s reward associated with purchases is once again attenuated. Since the midprice is frozen throughout the auction phase in our model, any inventory remaining after the auction is priced at the last exogenous midprice $S_{\tau^{\text{op}}}^{\text{mid}}$ rather than at the auction

clearing price. This separates the valuation of residual inventory from the endogenous auction price, which may be affected by the agent's own submitted schedules. The final reward of the market maker is given by

$$G(X_{\tau^{\text{cl}}}) = S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}} - \lambda |I_{\tau^{\text{cl}}}|^2 + f^a \left(S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}} \right) + S_{\tau^{\text{op}}}^{\text{mid}} I_{\tau^{\text{cl}}}.$$

Remark 3.7. *In the numerical part we will choose $f^c(t) = \min(1, \frac{t_+}{k^* \alpha})$ and $f^a(t) = q(-t)_+$ for some $q \in [0, 1]$. One can interpret the penalty as removing a fraction q of the reward. With $q = 1$, the purchase-side component is neutralized.*

To summarize, at time $t < \tau^{\text{cl}}$ the random one-step reward is

$$r_t(X_t, A_t) = \begin{cases} S_t^\bullet E_t f^c(k^* \alpha - (H_t^{\text{cl}} - S_t^\bullet)_+), & \text{if } t < \tau^{\text{op}}, \\ K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a) + f^a(K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a)) - d_t c_t, & \text{if } \tau^{\text{op}} \leq t < \tau^{\text{cl}}. \end{cases}$$

In our setting, the agent chooses action A_t in state X_t . Then, the executed volume E_t is realized, determining the one-step reward $r_t(X_t, A_t)$ during the state transition. Finally, the agent transitions into state X_{t+1} .

The learning objective of the agent is to solve

$$\begin{aligned} (\mathbf{P}) \quad & \sup_{\pi \in \Pi} J(\pi) = \mathbb{E}_\pi \left[\sum_{i=0}^m r_{t_i}(X_{t_i}, A_{t_i}) + G(X_{\tau^{\text{cl}}}) \right] \\ \text{s.t.} \quad & \begin{cases} X_0 \sim \mu_0 \\ A_{t_i} \sim \pi(\cdot \mid \varphi(X_{t_i})) \end{cases} \end{aligned}$$

For comparison of policies, we consider the expected economic performance $\bar{J}(\pi)$, which corresponds to the expected marked-to-market PnL, and is defined as

$$\forall \pi \in \Pi, \quad \bar{J}(\pi) = \mathbb{E}_\pi \left[\sum_{i=0}^n S_{t_i}^\bullet E_{t_i} + S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}} + S_{\tau^{\text{op}}}^{\text{mid}} I_{\tau^{\text{cl}}} - S_0^{\text{mid}} I_0 - \sum_{i=n+1}^m d_{t_i} c_{t_i} \right]$$

We similarly define $\bar{J}_\lambda(\pi) = \bar{J}(\pi) - \lambda \mathbb{E}_\pi [I_{\tau^{\text{cl}}}^2]$ the risk-adjusted expected marked-to-market PnL, for $\pi \in \Pi$. Without any reward shaping, we notice that $J(\pi) = \bar{J}_\lambda(\pi) + S_0^{\text{mid}} I_0$. The unshaped training criterion and the reported risk-adjusted economic criterion differ only by a constant, which unifies the base training objective with the economic objective. For precision, by the unshaped specification, we mean that the CLOB multiplier is set identically equal to one, the entire interim auction term $K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a) + f^a(K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a))$ is omitted, and the terminal f^a term is set to zero.

4 Learning Algorithms

In this section, we explain the numerical methods used to approximately solve $(\mathbf{P})$.

4.1 Deep Q-Network

We remind that $\mathcal{Y} = \varphi(\mathcal{X})$ is the feature space used by the learning algorithms. We consider the online episodic RL setting. In this setting, the agent executes the process sequentially for E episodes and we denote the total number of episode samples $N_s = \sum_{e=1}^E (m_e + 1)$. We consider an episodic, undiscounted, finite-horizon control problem. The transition law and conditional expected rewards of the simulator are not assumed known. We will therefore use a model-free method to train an approximately optimized feature-based policy $\hat{\pi}$ maximizing J . We approximate the action-value function by a neural network $Q_\theta(y, a)$ for some weight $\theta \in \mathbb{R}^q$, where $q \geq 1$. Our setting is however organized in two phases. For $y \in \mathcal{Y}$, let $t(y)$ be the time coordinate of y . We hence define a separate neural network for each phase:

$$\forall (y, a) \in \mathcal{Y} \times \mathcal{A}, \quad Q_\theta(y, a) = Q_\phi(y, a)\mathbf{1}_{\{t(y) < \tau^{\text{op}}\}} + Q_\psi(y, a)\mathbf{1}_{\{t(y) \geq \tau^{\text{op}}\}}$$

where $\theta = (\phi, \psi)$ and $\phi \in \mathbb{R}^{q_1}, \psi \in \mathbb{R}^{q_2}$ and $q_1 + q_2 = q$. The terminal Q-function is given by the reward $G(X_{\tau^{\text{cl}}})$ and we do not need to define a neural network. We train by Q-learning with experience replay and target networks (DQN, [52]) using the Double-DQN target of [61], and ensuring junction at $t = \tau^{\text{op}}$ when the phase switch occurs. Note that the ε -greedy training policy is stochastic, while the reported greedy evaluation policy is deterministic after applying a fixed tie-breaking rule.

4.2 Continuous-Action Control

The Deep Q-Network approach described above is adapted to the discrete action space of $(\mathbf{P})$. Discrete actions are naturally adapted for variables like a price, who is submitted according to a tick size anyway. For other variables however, like the supply/demand slope during the auction, it may be more effective to allow continuous actions. In this work, DQN is viewed as the discrete-RL baseline method and we want to compare it to different continuous-action methods. We hence consider a continuous relaxation of the action space. For each state $y \in \mathcal{Y}$, we introduce a relaxed continuous box of actions $\mathcal{A}^{\text{rel}}(y) \supseteq \text{Adm}(y)$. Before a relaxed action is “sent” to the environment, it is transformed into an admissible market making action through a projection $\Gamma_y : \mathcal{A}^{\text{rel}}(y) \rightarrow \text{Adm}(y)$. Informally, for $y \in \mathcal{Y}$ and an action $u \in \mathcal{A}^{\text{rel}}(y)$, the environment executes $a = \Gamma_y(u)$, the replay buffer stores u and the critic is trained on u . The precise componentwise projection Γ_y used in the numerical implementation is provided with the accompanying code. The continuous-control algorithms interact with the same constrained market making environment as DQN, but they optimize over a continuous parametrization of the action. The environment executes $\Gamma_y(u)$, while the critic is parameterized on the raw action u . Because Γ_y includes clipping, rounding and

Algorithm 2 Double DQN with phase-specific networks

Require: Episodes E , phase buffers $\mathcal{D}_c, \mathcal{D}_a$ of capacity M , warm-up size N_w , minibatch size B , reward scale c_r , Polyak coefficient τ_{targ} , exploration schedule $(\varepsilon_e)_{e \geq 1}$

- 1: Initialize $\theta_c = \phi$, $\theta_a = \psi$, $\theta_c^- = \theta_c$, $\theta_a^- = \theta_a$, and empty buffers $\mathcal{D}_c, \mathcal{D}_a$
- 2: **for** $e = 1, \dots, E$ **do**
- 3: Reset the environment, construct the realized grid $(t_i)_{0 \leq i \leq m_e+1}$ and observe X_{t_0}
- 4: **for** $i = 0, \dots, m_e$ **do**
- 5: Set $y_i \leftarrow \varphi(X_{t_i})$ and $\kappa_i \leftarrow c\mathbf{1}_{\{t_i < \tau^{\text{op}}\}} + a\mathbf{1}_{\{t_i \geq \tau^{\text{op}}\}}$
- 6: Select

$$a_i \sim \begin{cases} \mathcal{U}(\text{Adm}(y_i)), & \text{with probability } \varepsilon_e, \\ \arg \max_{a \in \text{Adm}(y_i)} Q_{\theta_{\kappa_i}}(y_i, a), & \text{with probability } 1 - \varepsilon_e \end{cases}$$
- 7: Apply a_i at t_i and observe the one-step reward r_i and the next simulator state $X_{t_{i+1}}$
- 8: Set $\eta_i \leftarrow \mathbf{1}_{\{i=m_e\}}$ and $g_i \leftarrow \eta_i G(X_{\tau^{\text{cl}}})$
- 9: If $\eta_i = 0$, set $y_{i+1} \leftarrow \varphi(X_{t_{i+1}})$; otherwise set $y_{i+1} \leftarrow \emptyset$
- 10: Store $(y_i, a_i, c_r r_i, y_{i+1}, \eta_i, c_r g_i)$ in $\mathcal{D}_{\kappa_i}$
- 11: **if** $|\mathcal{D}_{\kappa_i}| \geq N_w$ **then**
- 12: Sample $\mathcal{B} = (y_j, a_j, r_j, y'_j, \eta_j, g_j)_{1 \leq j \leq B}$ uniformly from $\mathcal{D}_{\kappa_i}$
- 13: **for** $j = 1, \dots, B$ **do**
- 14: **if** $\eta_j = 1$ **then**
- 15: Set $Y_j \leftarrow r_j + g_j$
- 16: **else**
- 17: Set $\kappa'_j = c\mathbf{1}_{\{t(y'_j) < \tau^{\text{op}}\}} + a\mathbf{1}_{\{t(y'_j) \geq \tau^{\text{op}}\}}$ and $a_j^* \in \arg \max_{a \in \text{Adm}(y'_j)} Q_{\theta_{\kappa'_j}}(y'_j, a)$
- 18: Set $Y_j \leftarrow r_j + Q_{\theta_{\kappa'_j}^-}(y'_j, a_j^*)$
- 19: **end if**
- 20: **end for**
- 21: Update θ_{κ_i} by one Adam step on $L_{\mathcal{B}}(\theta_{\kappa_i}) = \frac{1}{B} \sum_{j=1}^B \ell \left(Q_{\theta_{\kappa_i}}(y_j, a_j) - Y_j \right)$ where ℓ is the Huber loss
- 22: Update $\theta_{\kappa_i}^- \leftarrow (1 - \tau_{\text{targ}})\theta_{\kappa_i}^- + \tau_{\text{targ}}\theta_{\kappa_i}$
- 23: **end if**
- 24: **end for**
- 25: **end for**
- 26: **return** $\hat{\pi}(y) \in \arg \max_{a \in \text{Adm}(y)} Q_{\theta_{\kappa(y)}}(y, a)$ for $y \in \mathcal{Y}$ where $\kappa(y) = c\mathbf{1}_{\{t(y) < \tau^{\text{op}}\}} + a\mathbf{1}_{\{t(y) \geq \tau^{\text{op}}\}}$

a binary cancellation threshold, it is generally nonsmooth and many-to-one. We therefore do not invoke the standard smooth-action convergence interpretation of these algorithms.

We consider three off-policy actor-critic algorithms: Deep Deterministic Policy Gradient (DDPG), Twin Delayed DDPG (TD3), and Soft Actor-Critic (SAC), see respectively [42], [24], and [32]. DDPG uses a deterministic actor with exploration noise. TD3 adds twin critics, target-policy smoothing, and delayed actor updates. Finally, SAC uses a stochastic actor and entropy regularization. These methods are natural continuous-control counterparts to Q-learning and they all learn both the critic Q_θ and an actor $\mu_\omega: y \in \mathcal{Y} \mapsto \mu_\omega(y) \in \mathcal{A}^{\text{rel}}(y)$ with weights $\omega \in \mathbb{R}^l$ with $l \geq 1$ (such that $\Gamma_y(\mu_\omega(y)) \in \text{Adm}(y)$). Note that SAC's actor is stochastic, that is $\pi_\omega(\cdot | y) \in \mathcal{P}(\mathcal{A}^{\text{rel}}(y))$, and not a point map. The main difference with DQN is that the maximization over actions in the Bell-

man target is no longer computed by enumerating all admissible actions. Instead, the next action is produced by a learned policy. We refer to [3] for details on each of DDPG, TD3 and SAC and summarized in the following parameter table used for all numerical simulations. Each actor-critic method uses separate phase-specific actors, critics, target networks and replay buffers, and uses the auction critic for the continuation target at the phase boundary.

Feature	DQN	DDPG	TD3	SAC
<i>Shared</i>				
Replay capacity M		50,000		
Learning starts N_w		5,000		
Minibatch size B		128		
Optimizer		Adam		
Hidden layers (ReLU)		2×64		
Gradient-norm clip		1.0		
Reward scale c_r		10^{-3}		
Gradient steps per env. step		1		
<i>Learning rates</i>				
Critic	1.5×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
Actor	-	1×10^{-4}	3×10^{-4}	3×10^{-4}
Temperature	-	-	-	3×10^{-4}
<i>Critics and target networks</i>				
Critic loss	Huber	MSE	MSE	MSE
Number of critics	1	1	2	2
Polyak coefficient τ_{targ}	0.0025	0.0025	0.005	0.005
<i>Exploration</i>				
Behaviour policy	ε -greedy	Gaussian	Gaussian	Stochastic
Range	$1 \rightarrow 0.01$	-	-	-
Warm-up/decay (episodes)	100/600	-	-	-
Action-noise std	-	0.1	0.1	-
Target entropy	-	-	-	$-2/-3$
<i>TD3 target smoothing</i>				
Smoothing std	-	-	0.2	-
Smoothing clip	-	-	0.5	-
Policy delay	-	-	2	-

Table 1: Hyperparameters of the discrete (DQN) and continuous-action (DDPG, TD3, SAC) agents

5 Stylized Reference Benchmarks

We compare the performance of our DQN-learned policy and the continuous-control extensions against two different stylized reference benchmarks for optimal market making. Both benchmarks serve an illustrative purpose of closed-form approaches as we primarily compare the DQN policy

against the continuous-control extensions. The first benchmark is adapted from the optimal market making models of Avellaneda and Stoikov [4] later solved explicitly by Guéant, Lehalle and Fernandez-Tapia [31]. We adapt the market making model to liquidation only and without risk aversion for the continuous phase. We then suggest an approximation allowing a straightforward application to our discrete-time setup. The second benchmark is the time weighted average price policy for the continuous phase. We adopt the same heuristic liquidation rule for the auction phase. These benchmarks are external reference policies for optimal liquidation and are not required to belong to the learned policy class Π . A theoretical solution to the optimal market making problem on a closing auction is left for future work.

5.1 Avellaneda-Stoikov Optimal Market Making

In this section, we will recall the main results of [4] and [31], in the case where the market maker acts as a seller only. We consider the continuous phase, that is $t < \tau^{\text{op}}$ in what follows. Suppose we consider a continuous time setup, that is we work on $[0, t_n]$, where t_n is the last time of the continuous phase in our initial framework. The market mid price is assumed to follow an arithmetic Brownian motion $dS_t^{\text{mid}} = \sigma dB_t$ with $\sigma > 0$. We assume that transactions have constant size Δ . For simplicity, assume $\Delta = 1$. The inventory process writes $q_t = I_0 - N_t^a$, where N^a is the point process, independent of B , giving the cumulative number of shares sold by the market maker. Formulated initially by Avellaneda and Stoikov, we assume that the intensity of N^a depends on the spread $\delta_t^a = k_t^\bullet - k_t^{\text{mid}}$ as $\lambda^a(\delta^a) = Ae^{-\alpha k \delta^a}$ for $A, k > 0$. The cash process X_t of the market maker evolves according to $dX_t = (S_t^{\text{mid}} + \alpha \delta_t^a) dN_t^a$. Let $T = t_n$ and $\tilde{\mathcal{A}}$ be the set of bounded predictable processes. The market maker optimizes

$$(\mathbf{M}) \quad \sup_{\delta^a \in \tilde{\mathcal{A}}} \mathbb{E} \left[X_T + q_T S_T^{\text{mid}} \right]$$

Proposition 5.1. *The optimal quotes solving (M) are given by*

$$\delta^{a,*}(t, q) = \frac{1}{\alpha k} \left[1 + \ln \left(\frac{v_q(t)}{v_{q-1}(t)} \right) \right],$$

where

$$\forall q \in \llbracket 0, Q \rrbracket, \quad v_q(t) = \sum_{j=0}^q \frac{(Ae^{-1}(T-t))^j}{j!}.$$

Proof. Note that (M) corresponds to the risk-neutral market maker $\gamma \rightarrow 0$ in the standard optimal market making problem investigated in [4, 31]. In this case, one proves that the problem is reduced to find the solution to the ODE system

$$\forall q \in \llbracket 1, Q \rrbracket, \quad v'_q(t) = -Ae^{-1}v_{q-1}(t),$$

and we directly get optimal quotes from [31] when $\gamma \rightarrow 0$ by

$$\delta^{a,*}(t, q) = \frac{1}{\alpha k} \left[1 + \ln \left(\frac{v_q(t)}{v_{q-1}(t)} \right) \right].$$

We prove the result by induction. Note that $v_0 = 1$. Assume now that

$$\forall q \geq 1, \quad v_{q-1}(t) = \sum_{j=0}^{q-1} \frac{(Ae^{-1}(T-t))^j}{j!}.$$

We compute

$$\begin{aligned} v_q(t) &= v_q(T) - \int_t^T v'_q(s) ds = 1 + Ae^{-1} \int_t^T v_{q-1}(s) ds = 1 + Ae^{-1} \sum_{j=0}^{q-1} \int_t^T \frac{(Ae^{-1}(T-s))^j}{j!} ds \\ &= 1 + \sum_{j=0}^{q-1} \frac{(Ae^{-1})^{j+1}}{j!} \int_0^{T-t} s^j ds = 1 + \sum_{j=0}^{q-1} \frac{(Ae^{-1}(T-t))^{j+1}}{(j+1)!} = \sum_{j=0}^q \frac{(Ae^{-1}(T-t))^j}{j!}. \end{aligned}$$

This completes the proof by induction. $\square$

The optimal quotes $\delta^{a,*}(t, q)$ derived above hold in continuous time. These quotes yield, at time t and given the current inventory q , the optimal quote price (that is $S_t^{\text{mid}} + \alpha \delta^{a,*}(t, q)$). However, we work in discrete time in our setting. Therefore, if at the end of period t , when action (v_t, δ_t) has been submitted, m_t shares have been sold, it can be viewed as if on $[t, t+1)$, m_t/Δ fills occurred, all at price $\delta^{a,*}(t, q)$. The volume to be submitted is $v_t = q_t$ in this approximation. This allows to ensure that if enough market takers come to the market, execution is not limited by the volume exposed by the market taker. The approximation is twofold: each of the fills is at price $\delta^{a,*}(t, q_t)$ (instead of $\delta^{a,*}(s, q_s)$ for $t \leq s < t+1$) and we expose the whole current inventory at any time t . The notion of “exposed volume” does not exist in the continuous time setting because of the fixed transaction size. To conclude, the action of the market maker on the continuous phase writes $(V_{\max} \wedge \lfloor q_t \rfloor, \delta^{a,*}(t, 1 \vee \lfloor q_t \rfloor))$ such that it is admissible, where we implicitly project $\delta^{a,*}(t, 1 \vee \lfloor q_t \rfloor)$ itself on the set of admissible actions.

Remark 5.1. *Note that whenever $\delta^{a,*}(t, q_t)$ is not integer, we take the closest integer value instead.*

Once the auction opens, an inventory $q_{\tau^{\text{op}}} \geq 0$ remains. We implement the following heuristic policy. Let $\tilde{S}$ be the average between the mean and the max price of executed orders during the continuous phase. If no CLOB execution occurred, we set $\tilde{S} := H_{t_n}^{\text{cl}}$. The whole remaining inventory $q_{\tau^{\text{op}}}$ is put on the auction at $\tilde{S}$, with supply function $\tilde{g}_z(p) = \min(q_{\tau^{\text{op}}}, \min(zq_{\tau^{\text{op}}}, \beta K_{\max})(p - \tilde{S})_+)$. In Section 6, we consider $z = 10$. This single order is submitted right at the beginning of the auction, and only potentially executed at the clearing time. If $q_t = 0$, the benchmark submits no CLOB order; the quote formula is used only for $q_t \geq 1$.

5.2 Time-Weighted Average Price Benchmark

The second benchmark is deliberately simpler. Given the current inventory q_t , the trader submits a deterministic volume $v_t = V_{\max} \wedge \lfloor q_t \rfloor \wedge \lceil q_t / (T - t + 1) \rceil$, quoted at the best ask price, *i.e.* $\delta_t = 1$. This strategy corresponds to a uniform liquidation of the remaining inventory over the residual trading horizon. However, such a policy does not guarantee full liquidation during the continuous trading phase, as execution is conditional on order matching. For the auction phase, we adopt exactly the same liquidation policy as in the Avellaneda–Stoikov benchmark.

6 Numerical Simulations

This section employs the generative stochastic market model formulated in Section 2 to simulate continuous trading and closing auctions. We compare the DQN-learned policy against the continuous-control methods and the two stylized reference benchmarks: Avellaneda–Stoikov (AS) strategy, see Section 5.1 and the TWAP strategy see Section 5.2. The goal of this section is to emphasize the performance of the baseline DQN-learned policy compared with the continuous-control extended policies and the benchmarks. We start by generating an emulator of the CLOB followed by the closing auction. The algorithm used to generate the market mechanism is defined in Section 6.1 with Algorithm 3. In Section 6.3 we generate mid price process values by using the rough-Heston to describe the evolution of the price of a risky asset, see [28]. Finally, in Section 6.4 we train and test our DQN-learned policy to find the optimal trading strategy along a trading day with historical data for CAT, PG, GOOGL, JPM and MSFT. This second simulation is referred to as “historical” for simplicity. In both cases (generated or historical mid-price input) we note that our learning algorithm outperform the benchmarks on the mean returns.

In the synthetic simulations we consider a stylized high-frequency setting: a two-minute CLOB phase (120 nominal one-second steps) followed by a 30-second closing auction (30 steps), with the mid price generated by the rough Heston model presented in Section 6.3.1. In the simulations with historical mid-price input, each time step is a minute and we consider a two-hour CLOB phase followed by a 30-minute closing auction. This corresponds roughly to what happens at exchanges like Xetra and the CLOB phase can be viewed as the trading sessions leading up to the closing auction. We train the models on 1,000 (resp. 500) episodes in the synthetic (resp. “historical”) simulations and then run an additional 100 episodes keeping the policy fixed for evaluation. The networks receive normalized versions of physical time, decision index and continuous features, with normalization fitted on training data only. Throughout training, we consider a validation set which allows to select the report policy by early stopping.

6.1 Generative Stochastic Market Model

We now explain in details our market emulator to generate the financial market in our setting. The algorithm is described in Algorithm 3. It specifies only the exogenous random variables and

accepted exogenous book updates. Let us precise that we say that a candidate exogenous linear auction book is valid if the aggregate slope $G_t \geq D_\mu$ for a numerical lower slope bound $D_\mu > 0$ and there is a nonnegative continuous clearing root, as in Proposition 2.1.

At each auction decision time, the accepted exogenous update returned by Algorithm 3 is incorporated into the current configuration before the agent acts via the following chronology.

1. If $k = n + 1$, set $H_{t_{n+1}}^{\text{cl}} = H_{t_n}^{\text{cl}}$. If $k \geq n + 2$, compute $H_{t_k}^{\text{cl}}$ from the complete post-action book stored at t_{k-1} .
2. Sample the exogenous proposals at t_k and accept each proposal only if the resulting exogenous book remains valid.
3. Construct the pre-action configuration X_{t_k} and reveal $\varphi(X_{t_k})$ to the agent.
4. If $c_{t_k} = 1$, remove every still-active agent schedule submitted strictly before t_k .
5. Add the current schedule $p \mapsto K_{t_k}^a(p - S_{\tau^{\text{op}}}^{\text{mid}} - \alpha b_{t_k}^a)$.
6. Store the resulting post-action book. If $k = m$, freeze this book, process no event on $(t_m, \tau^{\text{cl}}]$, and apply the terminal clearing and allocation rule.

After the final action at t_m , the controlled environment freezes the post-action book and applies the clearing and allocation rule of Section 2.1.3.

The inputs of our emulator are given by $(\lambda_0, v_m, \gamma_m, V, \beta_a, \beta_b, V_\infty, \rho, U_1, U_2, M_1, M_2, p_1, p_2, p_3, p_4)$.

- λ_0 represents the intensity of the counting processes N^+ and N^- on the continuous phase, modeled as independent Poisson processes. The two processes are not observable for the agent during the continuous phase, but only during the auction phase. They are sampled in the first step of the algorithm.
- (v_m, γ_m) are the parameters of the Pareto distribution modeling the volumes of the limit orders throughout the whole trading sessions. This reproduces the well-known heavy tail behavior of the density of market order size, see for example [29, 25, 10]. The market model allows orders of maximum size V : greater liquidity takers will accumulate at volume V .
- (β_a, β_b) represent the parameters of the Beta distribution modeling the first volume of the limit order book. It is similar to the Beta scaling effect described in for example [35, 63]. Note that V_∞ is a rescaling parameter as the Beta distribution has support $[0, 1]$. Further volumes in the limit order book decay geometrically with parameter $\rho \in (0, 1]$ and are truncated via the maximum order book depth L_{max} , that is $V_t^{\zeta, j} = 0$ for $j > L_{\text{max}}$.
- (U_1, U_2) represent the bounds between which exogenous market maker sample their supply slopes from during the auction phase.

Algorithm 3 Exogenous stochastic market generator

Require: Parameter vector $(\lambda_0, v_m, \gamma_m, V, \beta_a, \beta_b, V_\infty, \rho, U_1, U_2, M_1, M_2, p_1, p_2, p_3, p_4)$

- 1: Sample the independent CLOB counting processes $N^+, N^- \sim \text{PP}(\lambda_0)$ on $[0, \tau^{\text{op}}]$
- 2: Construct the complete realized grid $(t_i)_{0 \leq i \leq m+1}$ according to Definition 3.1
- 3: **for** $k = 0, \dots, n$ **do**
- 4: **for** $\zeta \in \{+, -\}$ **do**
- 5: Sample $V_{t_k}^{\zeta, 1} \sim V_\infty \text{Beta}(\beta_a, \beta_b)$
- 6: **for** $\ell = 1, \dots, L_{\max} - 1$ **do**
- 7: Set $V_{t_k}^{\zeta, \ell+1} = \rho V_{t_k}^{\zeta, \ell}$
- 8: **end for**
- 9: **for** $r = N_{t_k}^\zeta + 1, \dots, N_{t_{k+1}}^\zeta$ **do**
- 10: Sample $Z_{k,r}^\zeta \sim \text{Pareto}(v_m, \gamma_m)$ and set $\nu_{k,r}^\zeta = \min\{Z_{k,r}^\zeta, V\}$
- 11: **end for**
- 12: **end for**
- 13: Send exogenous CLOB snapshot and sampled arrivals on $(t_k, t_{k+1}]$ to the controlled env
- 14: **end for**
- 15: Receive from the controlled environment the post-execution residual exogenous CLOB multiset $O_{\tau^{\text{op}}-}^*$, recompute the final Algorithm 1 calibration, and initialize the carry-over schedules calibration from this residual multiset and initialize the carry-over schedules

$$\mathcal{B}_{\tau^{\text{op}}}^* = \{g_{k, \tau^{\text{op}}}^* : k \in \mathcal{K}_*\}$$

- 16: Initialize the exogenous auction book from the carry-over book and set $H_{t_{n+1}}^{\text{cl}} \leftarrow H_{t_n}^{\text{cl}}$
 - 17: **for** $k = n + 1, \dots, m$ **do**
 - 18: Sample proposed auction updates $B_{t_k} \sim \mathcal{B}(p_1)$, $D_{t_k} \sim \mathcal{B}(p_2)$, $J_{t_k}^\pm \sim \mathcal{B}(p_3)$, $G_{t_k}^\pm \sim \mathcal{B}(p_4)$
 - 19: **if** $B_{t_k} = 1$ **then**
 - 20: Sample $K_{t_k}^r \sim \mathcal{U}([U_1, U_2])$, $S_{t_k}^r \sim S_{\tau^{\text{op}}}^{\text{mid}} + \alpha \mathcal{U}(\llbracket M_1, M_2 \rrbracket)$ and add the new exogenous schedule
 - 21: **end if**
 - 22: **if** $D_{t_k} = 1$ and a nonpersistent exogenous schedule is present **then**
 - 23: Propose canceling one uniformly selected exogenous schedule and accept the cancellation only if the resulting exogenous book remains valid
 - 24: **end if**
 - 25: **for** $\zeta \in \{+, -\}$ **do**
 - 26: **if** $J_{t_k}^\zeta = 1$ **then**
 - 27: Sample $Z_{t_k, \zeta} \sim \text{Pareto}(v_m, \gamma_m)$ and set $\nu_{t_k, \zeta}^\zeta = \min\{Z_{t_k, \zeta}, V\}$
 - 28: Add the corresponding market order if the resulting book is valid
 - 29: **end if**
 - 30: **if** $G_{t_k}^\zeta = 1$ and an active market order is present on side ζ **then**
 - 31: Cancel a uniformly sampled order if the resulting book is valid
 - 32: **end if**
 - 33: **end for**
 - 34: **if** $k = m$ **then**
 - 35: Process no further exogenous update
 - 36: **end if**
 - 37: **end for**
-

- (M_1, M_2) represent the integer bounds of the price at which exogenous market makers quote during the auction phase. Exogenous market makers during the volume are assumed to be sampled as $S_t^i \sim S_{\tau_{\text{op}}}^{\text{mid}} + \alpha \mathcal{U}(\llbracket M_1, M_2 \rrbracket)$ because the auction opening time is the last time market participants see the mid price. Note that this shape of auction price have been introduced in [20].
- (p_1, p_2, p_3, p_4) represent the independent proposal probabilities that drive the market structure during the auction phase. A new market maker arrives per step with proposal probability p_1 , one market maker cancels his order with proposal probability p_2 , a market taker arrives on either side with proposal probability p_3 , and a market taker cancels his order with proposal probability p_4 , all the conditioned on the book being still valid. These events are sampled using a Bernoulli distribution.

The proposal variables and their marks are sampled independently. The accepted book updates are state-dependent because of the validity screen. We do assume that the limit order book is refreshed from one time step to the next one. Lines 3 to 6 describe the mechanics of the continuous phase. Lines 7 to 24 describe the mechanics of the auction phase. Finally, we precise that all sampled quantities are projected on the grid we fixed in Section 3.

6.2 Benchmark Simulations

We start by simulating the optimal strategies for the Benchmarks in Figure 1, which corresponds to the synthetic setting. On the top left, we show the evolution of the inventories for both AS and TWAP, on the top right we simulate the continuous phase limit price, on the bottom left the optimal volume submitted during the continuous phase, and on the bottom right we show the optimal slope K_t during the auction. Note that the TWAP fails to liquidate all the inventory before the auction starts unlike the AS strategy explaining the difference of order executed in the auction between these two benchmarks.

6.3 Rough Heston Model for the Mid Price: Generative Data Approach

In this section, we generate data from the rough-Heston model. We consider a trading session with a 2 minute continuous phase, followed by a 30 second closing auction.

6.3.1 Numerical Method and Parameter Calibration

In the first numerical implementation, we assume that the mid price S_t follows a rough Heston model. The motivation is based on the so-called rough volatility of financial assets [28, 12, 2]. Consider $\rho \in [-1, 1]$ a constant, $S_0 = 100$ (which is the numerical value we work with in this

Symbol	Value	Comment
τ^{op}	120	Auction opening time
τ^{cl}	150	Clearing time
I_0	100	Initial inventory
λ_0	1	Continuous phase Poisson intensity
v_m	2	Pareto distribution scale parameter
γ_m	2.5	Pareto distribution shape parameter
V_∞	15	Beta distribution scaling parameter
β_a	2	First Beta distribution shape parameter
β_b	5	Second Beta distribution shape parameter
ρ	0.5	Limit order book volume decay parameter
V_{max}	30	Maximum volume admitted by the market
L_{max}	12	Maximum LOB depth and agent quote
B_{max}	25	Maximum deviation for price during auction
D_μ	0.1	Numerical lower slope bound
U_1	0.1	Exogenous supply slope lower bound
U_2	2	Exogenous supply slope upper bound
M_1	-10	Exogenous supply spread lower bound
M_2	10	Exogenous supply spread upper bound
p_1	1	New market maker arrival probability
p_2	0	Market maker cancellation probability
p_3	0.3	New market taker arrival probability
p_4	0.05	Market taker cancellation probability
λ	0.5	Inventory penalty
q	1	Wrong-side dealing penalty
k^*	1000	Tolerance
d	0.1	Cancellation cost per unit
α	0.01	Tick size
β	10/3	Tick size of grid on K^a
K_{max}	10	Upper bound on K^a/β

Table 2: Generative stochastic market model parameters

section), V_0 , H , θ , ς and ν be positive constants. Recalling [28, 57], the rough Heston model writes

$$\begin{cases} S_t &= S_0 + \int_0^t S_s \sqrt{V_s} d(\rho B_s + \sqrt{1 - \rho^2} B_s^\perp), \\ V_t &= V_0 + \int_0^t K_H(t-s)((\theta - \varsigma V_s)ds + \nu \sqrt{V_s} dB_s), \end{cases}$$

where $(B, B^\perp)$ are two independent Brownian motions and $K_H(t) = t^{H-1/2}/\Gamma(H+1/2)$ for $t > 0$.

We used the Euler-type scheme of [57] to approximate this rough Heston model. For each episode, we first construct the realized CLOB grid $0 = t_0 < \dots < t_n = \tau^{\text{op}} - 1 < t_{n+1} = \tau^{\text{op}}$. Conditional on this complete grid, and independently of the order-flow processes, we simulate the rough-Heston approximation on the resulting nonuniform mesh. We use the model time measured in trading years. Thus, for $s^* = 252 \cdot 6.5 \cdot 3,600$, let $\bar{t}_i = t_i/s^*$. We set $\Delta \bar{t}_{i+1}^n = \bar{t}_{i+1} - \bar{t}_i$ for $i \in \llbracket 0, n \rrbracket$ and generate the

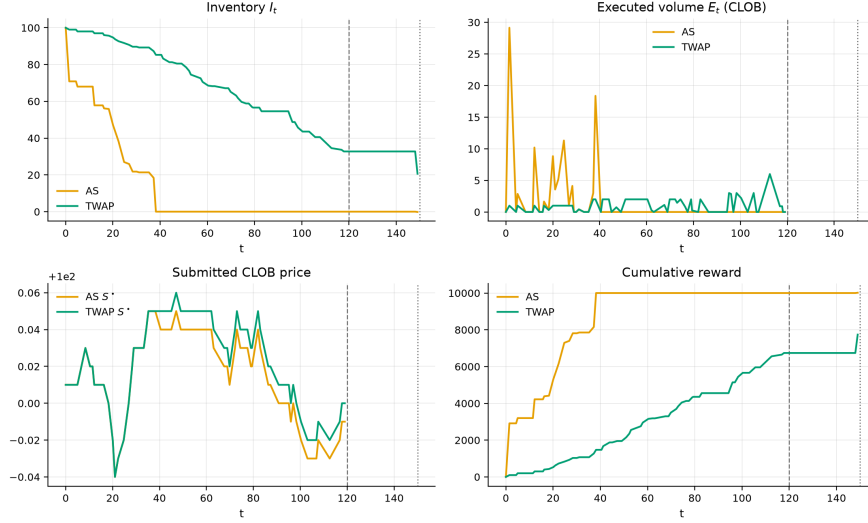


Figure 1: Benchmark strategies on one evaluation episode (AS and TWAP) in the synthetic setting for fixed seed

Brownian increments with variance $\Delta \bar{t}_{i+1}^n$. The midprice is simulated through $t_{n+1} = \tau^{\text{op}}$, projected to the tick grid, and then frozen throughout the auction. Then $S_k = \exp(Y_k)$ for $k \in \llbracket 0, n+1 \rrbracket$ where

$$\begin{cases} Y_k &= Y_0 + \sum_{i=0}^{k-1} \left(-\frac{1}{2}(V_i)_+ \Delta \bar{t}_{i+1}^n + \rho \sqrt{(V_i)_+} (B_{\bar{t}_{i+1}} - B_{\bar{t}_i}) + \sqrt{1-\rho^2} \sqrt{(V_i)_+} (B_{\bar{t}_{i+1}}^\perp - B_{\bar{t}_i}^\perp) \right), \\ V_k &= V_0 + \sum_{i=0}^{k-1} \left(K_H (\bar{t}_k - \bar{t}_i) (\theta - \varsigma (V_i)_+) \Delta \bar{t}_{i+1}^n + K_H (\bar{t}_k - \bar{t}_i) \nu \sqrt{(V_i)_+} (B_{\bar{t}_{i+1}} - B_{\bar{t}_i}) \right). \end{cases}$$

At time t_k , the midprice revealed to the trading environment is $S_{t_k}^{\text{mid}} = \alpha \lfloor S_k / \alpha + 1/2 \rfloor$. In numerical applications, we used the illustrative parameters $H = 0.1$, $\rho = -0.7$, $V_0 = 0.02$, $\theta = 0.02$, $\varsigma = 0.3$ and $\nu = 0.3$ adapted from [1].

Now, one may notice that the AS stylized reference benchmark assumes that $dS_t = \sigma dB_t$, instead of the rough Heston model. In this sense, the goal of the numerical simulation is to compare how our deep reinforcement learning policies performs against a stylized reference benchmark, without claiming optimality of this benchmark. The benchmark is deliberately stylized and is not optimal for the simulated rough-Heston environment; it is therefore interpreted as a reference policy rather than as an oracle comparator.

We calibrated the value σ by estimating the standard deviation of the mid price on the trading period. From [4], we have $A = \lambda_0 / \gamma_m$ and $k = \alpha K$. Here, K is the proportionality constant in the empirical relationship $K \Delta p = \ln(Q)$, where Δp is the move in price when a market order of size Q arrives. We did a least squares regression to determine K , by simulating 5,000 samples of the limit order book and market orders $Q \sim \text{Pareto}(v_m, \gamma_m)$.

6.4 Simulation with Historical Mid-Price Input

7 Conclusion

We formulate a two-phase liquidation problem that combines one-sided CLOB execution with signed participation in a stylized closing call, and develop feature-based reinforcement-learning methods for the resulting controlled simulator. Our findings suggest that RL has the potential to be effective for market making in complex structures beyond simple LOBs with closing auction, as for example workup session or AHEAD mechanism [22, 19] or sequence of periodic auctions [49, 43, 11].

The model is a single-agent best-response problem against fixed stochastic background policies; it does not model strategic responses or market equilibrium. The auction chronology is sequential and includes a final reaction opportunity for the agent. The background CLOB is refreshed between decisions, and the carry-over is a calibrated schedule representation rather than a literal order transfer. The reduced feature process need not be Markov. Negative terminal inventory is permitted, but post-close securities-borrow and financing costs are not modeled explicitly; terminal exposure is instead marked to the frozen midprice and penalized quadratically. Historical-input experiments retain synthetic order flow and auction dynamics.

References

- [1] Eduardo Abi Jaber. Lifting the Heston model. *Quantitative finance*, 19(12):1995–2013, 2019.
- [2] Eduardo Abi Jaber and Omar El Euch. Multifactor approximation of rough volatility models. *SIAM journal on financial mathematics*, 10(2):309–349, 2019.
- [3] Josh Achiam. Spinning up in deep reinforcement learning. <https://spinningup.openai.com/>, 2018. OpenAI documentation; accessed 18 June 2026.
- [4] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.
- [5] Bastien Baldacci, Philippe Bergault, and Olivier Guéant. Algorithmic market making for options. *Quantitative Finance*, 21(1):85–97, 2021.
- [6] Mario Bellia, Loriana Pelizzon, Marti G Subrahmanyam, and Darya Yuferova. Market liquidity and competition among designated market makers. *Management Science*, 71(1):184–201, 2025.
- [7] Taweh Beysolow II. Market making via reinforcement learning. In *Applied Reinforcement Learning with Python: With OpenAI Gym, Tensorflow, and Keras*, pages 77–94. Springer, 2019.
- [8] Bruno Biais and Anne Marie Faugeron-Crouzet. IPO auctions: English, dutch,... french, and internet. *Journal of Financial Intermediation*, 11(1):9–36, 2002.
- [9] Bruno Biais, Pierre Hillion, and Chester Spatt. Price discovery and learning during the pre-opening period in the paris bourse. *Journal of Political Economy*, 107(6):1218–1248, 1999.
- [10] Jean-Philippe Bouchaud. Power laws in economics and finance: some ideas from physics. *Quantitative finance*, 1(1):105, 2001.
- [11] Eric Budish, Peter Cramton, and John Shim. Implementation details for frequent batch auctions: Slowing down markets to the blink of an eye. *American Economic Review*, 104(5):418–424, 2014.
- [12] Nicole Bäuerle and Sascha Desmettre. Portfolio optimization in fractional and rough Heston models. *SIAM Journal on Financial Mathematics*, 11(1):240–273, 2020.
- [13] Jialun Cao, David Šiška, Lukasz Szpruch, and Tanut Treetanthiploet. Logarithmic regret in the ergodic Avellaneda-Stoikov market making model. *arXiv preprint arXiv:2409.02025*, 2024.
- [14] Agostino Capponi and Charles-Albert Lehalle. *Machine Learning and Data Sciences for Financial Markets: A Guide to Contemporary Practices*. Cambridge University Press, 2023.
- [15] René Carmona. The influence of economic research on financial mathematics: Evidence from the last 25 years. *Finance and Stochastics*, 26(1):85–101, 2022.

- [16] Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [17] Nicolò Cesa-Bianchi, Tommaso Cesari, Roberto Colomboni, Luigi Foscari, and Vinayak Pathak. Market making without regret. *arXiv preprint arXiv:2411.13993*, 2024.
- [18] Christa Cuchiero, Hu Ruimeng, Sara Svaluto-Ferro, Xu Renyuan, et al. Special issue on machine learning in finance. *Mathematical Finance*, 34(2):259–261, 2024.
- [19] Joffrey Derchu, Philippe Guillot, Thibaut Mastrolia, and Mathieu Rosenbaum. AHEAD: Ad hoc electronic auction design. *Frontiers of Mathematical Finance*, 3(2):163–213, 2024.
- [20] Joffrey Derchu, Dimitrios Kavvathas, Thibaut Mastrolia, and Mathieu Rosenbaum. Equilibria and incentives for illiquid auction markets. *arXiv preprint arXiv:2307.15805*, to appear in *Market Microstructure and Liquidity*, 2023.
- [21] Songzi Du and Haoxiang Zhu. Welfare and optimal trading frequency in dynamic double auctions. Technical report, National Bureau of Economic Research, 2014.
- [22] Darrell Duffie and Haoxiang Zhu. Size discovery. *The Review of Financial Studies*, 30(4):1095–1150, 2017.
- [23] Jianqing Fan, Zhaoran Wang, Yuchen Xie, and Zhuoran Yang. A theoretical analysis of deep Q-learning. In *Learning for dynamics and control*, pages 486–489. PMLR, 2020.
- [24] Scott Fujimoto, Herke Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International conference on machine learning*, pages 1587–1596. PMLR, 2018.
- [25] Xavier Gabaix. Power laws in economics and finance. *Annu. Rev. Econ.*, 1(1):255–294, 2009.
- [26] Sumitra Ganesh, Nelson Vadori, Mengda Xu, Hua Zheng, Prashant Reddy, and Manuela Veloso. Reinforcement learning for market making in a multi-agent dealer market. *arXiv preprint arXiv:1911.05892*, 2019.
- [27] Bruno Gašperov and Zvonko Kostanjčar. Market making with signals through deep reinforcement learning. *IEEE access*, 9:61611–61622, 2021.
- [28] Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. In *Commodities*, pages 659–690. Chapman and Hall/CRC, 2022.
- [29] Parameswaran Gopikrishnan, Vasiliki Plerou, Xavier Gabaix, and H Eugene Stanley. Statistical properties of share volume traded in financial markets. *Physical review e*, 62(4):R4493, 2000.
- [30] Olivier Guéant and Iuliia Manziuk. Deep reinforcement learning for market making in corporate bonds: beating the curse of dimensionality. *Applied Mathematical Finance*, 26(5):387–452, 2019.

- [31] Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and financial economics*, 7(4):477–507, 2013.
- [32] Tuomas Haarnoja, Aurick Zhou, Kristian Hartikainen, George Tucker, Sehoon Ha, Jie Tan, Vikash Kumar, Henry Zhu, Abhishek Gupta, Pieter Abbeel, et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018.
- [33] Ben Hambly, Renyuan Xu, and Huining Yang. Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503, 2023.
- [34] Simon Jantschgi. Transaction cost (in) transparency: Coasian dynamics in frequent batch auctions. *Available at SSRN 4861066*, 2024.
- [35] Joseph Jerome, Gregory Palmer, and Rahul Savani. Market making with scaled beta policies. In *Proceedings of the Third ACM International Conference on AI in Finance*, pages 214–222, 2022.
- [36] Joseph Jerome, Leandro Sánchez-Betancourt, Rahul Savani, and Martin Herdegen. Mbt-gym: Reinforcement learning for model-based limit order book trading. In *Proceedings of the Fourth ACM International Conference on AI in Finance*, pages 619–627, 2023.
- [37] Paul Jusselin, Thibaut Mastrolia, and Mathieu Rosenbaum. Optimal auction duration: A price formation viewpoint. *Operations Research*, 69(6):1734–1745, 2021.
- [38] Eugene Kandel, Barbara Rindi, and Luisella Bosetti. The effect of a closing call auction on market quality and trading strategies. *Journal of Financial Intermediation*, 21(1):23–49, 2012.
- [39] Michael Kearns and Yuriy Nevmyvaka. Machine learning for market microstructure and high frequency trading. *High frequency trading: New realities for traders, markets, and regulators*, 72:1877–1901, 2013.
- [40] Albert S Kyle. Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, pages 1315–1335, 1985.
- [41] Laura Leal, Mathieu Laurière, and C-A Lehalle. Learning a functional control for high-frequency finance. *Quantitative Finance*, 22(11):1973–1987, 2022.
- [42] Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning, 2015.
- [43] Ananth Madhavan. Trading mechanisms in securities markets. *the Journal of Finance*, 47(2):607–641, 1992.

- [44] Ananth Madhavan and Venkatesh Panchapagesan. Price discovery in auction markets: A look inside the black box. *The Review of Financial Studies*, 13(3):627–658, 2000.
- [45] Thibaut Mastrolia and Tianrui Xu. Clearing time randomization and transaction fees for auction market design. *arXiv preprint arXiv:2405.09764*, 2024.
- [46] Thibaut Mastrolia and Tianrui Xu. Optimal rebate design: Incentives, competition and efficiency in auction markets. *arXiv preprint arXiv:2501.12591*, 2025.
- [47] Paul Milgrom. Auctions and bidding: A primer. *Journal of economic perspectives*, 3(3):3–22, 1989.
- [48] Paul Milgrom. Auction market design: Recent innovations. *Annual Review of Economics*, 11(1):383–405, 2019.
- [49] Paul R Milgrom. The economics of competitive bidding: a selective survey. *Social goals and social organization: Essays in memory of Elisha Pazner*, pages 261–292, 1985.
- [50] Paul Robert Milgrom. *Putting auction theory to work*. Cambridge University Press, 2004.
- [51] Volodymyr Mnih. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.
- [52] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *nature*, 518(7540):529–533, 2015.
- [53] Ralph Neuneier. Enhancing Q-learning for optimal asset allocation. *Advances in neural information processing systems*, 10, 1997.
- [54] Yuriy Nevmyvaka, Yi Feng, and Michael Kearns. Reinforcement learning for optimized trade execution. In *Proceedings of the 23rd international conference on Machine learning*, pages 673–680, 2006.
- [55] Brian Ning, Franco Ho Ting Lin, and Sebastian Jaimungal. Double deep Q-learning for optimal execution. *Applied Mathematical Finance*, 28(4):361–380, 2021.
- [56] Franck Raillon. The growing importance of the closing auction in share trading volumes. *Journal of Securities Operations & Custody*, 12(2):135–152, 2020.
- [57] Alexandre Richard, Xiaolu Tan, and Fan Yang. On the discrete-time simulation of the rough heston model. *SIAM Journal on Financial Mathematics*, 14(1):223–249, 2023.
- [58] Mohammed Salek, Damien Challet, and Ioane Muni Toke. Equity auction dynamics: latent liquidity models with activity acceleration. *Quantitative Finance*, 24(10):1381–1398, 2024.

- [59] Justin A Sirignano. Deep learning for limit order books. *Quantitative Finance*, 19(4):549–570, 2019.
- [60] Richard S Sutton, Andrew G Barto, et al. *Reinforcement learning: An introduction*, volume 1 of 1. MIT Press, 1998.
- [61] Hado Van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double q-learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 30 of 1, 2016.
- [62] Gustavo Vulcano, Garrett Van Ryzin, and Costis Maglaras. Optimal dynamic auctions for revenue management. *Management Science*, 48(11):1388–1407, 2002.
- [63] Yongzhao Wang, Rahul Savani, Anri Gu, Chris Mascioli, Theodore Turocy, and Michael Wellman. Market making with learned beta policies. In *Proceedings of the 5th ACM International Conference on AI in Finance*, pages 643–651, 2024.
- [64] Christopher JCH Watkins and Peter Dayan. Q-learning. *Machine learning*, 8(3):279–292, 1992.